

The alert model based on neural networks is used to retrieve, in real time, content from social networks (Twitter and Facebook). Once cleaned of duplicate and replicated content, we want to learn from the first-hand content, through manually tagged information, relevant content to warn and alert people and claims handlers to make quick and effective decisions, that will save lives. It is enhanced as a Machine Learning Environment (ALE) based on Deep Learning to fetch content from all over the Web. It is also enhanced to be a hybrid of a deep CNN-LSTM based ALE using awareness, assessment and education. Experiments have shown that it continues to give better results than previous work.



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Social Networks for Disaster Management

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Table of Abbreviations

Abbreviations	Meaning
AI	Artificial Intelligence
AIDR	Artificial Intelligence for Disaster Response
ANN	Automated Neural Network
API	Application Programming Interface
ARG	Alternative Realistic Game
BLSTM	Bi-Directional LSTM
BRNN	Bi-Directional RNN
CNN	Convolutional Neural Network
CNN-LSTM	Hybrid of Deep Convolutional NN and LSTM
ConvNets	Convolutional Neural Network
DM	Disaster Management
DL	Deep Learning
DOSN	Decentralized Online Social Network
FBNN	FeedBackward eural Network
FCLSTM	Fully Connected LSTM
FFNN	FeedForwark Neural Network
GAN	Generative Adversial Networks
GBED	Games based-Evacuation Drill
GIS	Geographic Information System
GPS	Global Positioning System
GSM	Global System for Mobile
HMD	Head-Mounted Display
IDS	Intrusion-Detection System
KDD	Knowledge Discovery in Databases
LR	Logic Regression
LSTM	Long Short-Term Memory
MDM	Motion Danger Map
MHM	Motion Hazard Map
ML	Machine Learning
MVC	Model-View-Controller
NB	Naive Bayes

Abbreviations	Meaning
NL	Neural Learning
NN	Neural Network
NSM	Non-Structural measures of Disaster Management
OCHA	Office of the Coordination of Humanitarian Affairs
OSN	Online Social Network
OSM	Open Street Map
P2P	Peer-to-Peer
POS	Par-of-Speech
PT	Penumbral Tourism
RBC	Rule-based Classifier
RBF	Radial Basis Function NN
RBM	Restricted Boltzman Machine
ResNet	Residual Neural Network
RF	Random Forest
RNN	Recurrent Neural Network
SM	Structural measures of Disaster Management
SNS	Social Network Site
SR	Symbolic Regression
SVM	Support Vector Machine
TED	Tsunami Evacuation Drill

Abstract

Nowadays, technology is constantly evolving and more and more quickly. An event is discussed in Social networking. It must ensure, before deployment, that an first-hand message will always arrive to disaster managers, to take quick and efficient decisions that will save lives, and this nonobstant its fake news. The real-time retrieving of relevant content from social media in crisis event is very important. This relentless development responds to the demand of the general public. Obviously, the integration of new functions, always more sophisticated, in our daily tools, makes their development more and more complex, in particular at the real time level.

Keywords : Disaster, Neural learning, Relevant content, Social media.

Résumé

De nos jours, la technologie évolue constamment et de plus en plus rapidement. Un événement est discuté dans les réseaux sociaux. Il doit s'assurer, avant le déploiement, qu'un message de première main arrivera toujours aux gestionnaires de catastrophes, pour prendre des décisions rapides et efficaces qui sauveront des vies, et ce nonobstant son fausses nouvelles. La récupération en temps réel du contenu pertinent des médias sociaux en cas de crise est très importante. Ce développement incessant répond à la demande du grand public. Evidemment, l'intégration de nouvelles fonctions, toujours plus sophistiquées, dans nos outils quotidiens, rend leur développement de plus en plus complexe, notamment au niveau temps réel.

Mots-Clés : Catastrophe, Apprentissage neuronal, Contenu pertinent, Média sociaux.

General Introduction

Introduction

In recent years, the frequency of large-scale disasters has increased worldwide. The Boumerdes earthquake [1] of May 21, 2003 and the Chlef earthquake (Algeria), for example, which occurred in 1980 (7.3 degrees on the Richter scale) caused 2,633 deaths and caused the collapse of 20,000 homes. Every time a disaster occurs, many recognize the importance of disaster management. The majority of disasters have huge and lasting impacts on human lives and societies. Disasters such as earthquakes, floods, explosions, terrorist acts and tsunamis result in catastrophic human suffering, loss of property and other negative consequences [2]. In order to minimize the consequences of such events, it is essential to prepare and develop resilient communities by educating them with the appropriate and up-to-date information needed to cope with disasters. We will examine here how individuals, organizations and communities develop their skills in response to unforeseen events, whether in preparation, in disaster management or in the institutional aspect. Our analytical framework is guided by the important social phenomenon that occurs in almost every disaster and the vastness of activity that brings this observation of evidence to wonder. Indeed, at each event, harmless it may be, people come running to share it with other people, particularly those who are connected. May this event be happy and even more so if it is an unfortunate event.

Motivations

People use social networking to post situational updates in various forms such as text messages, images and videos. Numerous studies [3] have shown that this online information is useful for a quick response to disaster management. These analyzes of the use of social networking in disasters have identified a distinct role for local users of the event (or "in the field"), who are more likely to generate useful information to improve situational awareness [4]. Thanks to the improvement of storage technics (big data) and the ease of use, the participatory culture and the interoperability of end-users (Web 2.0), we are being confronted with information overload stored, notably

the Web which is a popular and interactive way of disseminating information today. It is notably immense, diverse and dynamic. This new era is defined by the rapid development of knowledge and that traditional learning initiatives have not been able to cope with the growing complexity and rapid changes of the new knowledge society. Therefore, a new vision of learning is needed, namely Automated Learning. Its aim is to discover, in large amounts of data, valuable information that can help understand these data or predict the behavior of future data. This process involves examining large amounts of data (stored in warehouses or, notably, in streaming) using statistical, mathematical or eventually AI techniques. This knowledge, unknown at first, may be correlations, patterns or general trends in that data.

Methodology

Automated Learning is based on the idea of analyzing problems to understand their principles and develop appropriate mathematical models. For that, experimental data are required to verify the correction of the system or the estimation of some difficult parameters to mathematical modeling [5]. A fundamental shift towards a more open and learner-centered learning model is needed where Automated Learning uses various manual and automatic tools : we start by describing the data, summarize their statistical attributes (means, variances, covariance), visualize them using the curves, the graphs, the diagrams, and finally look for the significant links potential between variables (such as values that are repeated). But the description of the data alone does not provide a plan of action. We must build a prediction model based on the information discovered, and we validate this model on other data. Over the past decade, it has been argued that Automated Learning could meet the needs of the new knowledge society and transform learning.

However, in the majority of cases, systems do not have principles understood or are too complex for mathematical modeling. Automated Learning, which suggests complex and unimaginable specific impacts on the society and more broadly the economy, remains primarily in the field of computer science, but is closely linked to cognitive science, neuroscience, biology and psychology and could lead to the crossroads of systems, nanotechnology, biotechnology and cognitive science, leading to artificial intelligence with a wider base.

Smart Disaster education plays a key role in encouraging community members to develop disaster management skills. In addition, these members also share their knowledge and help one another in disaster preparedness, like with other disaster education techniques, namely, Evacuation drills, Games Based-Evacuation Drill [6], Paradigmatic Tourism [7], Dark Tourism, Penumbral Tourism, Mixed Realistic Games or Alternative

Realistic Games (ARG) [8] and tower defense game [9], or other evacuation techniques as Simulations.

In this research work, we present the theoretical, design, implementation and evaluation details of an environment, a learning framework whose main objective is to help users become familiar with all of these different learning models using a wide variety of media and digital data available in this environment. This automated learning environment is fundamentally personal, social, distributed, ubiquitous, flexible and dynamic.

Organisation of the thesis

This manuscript is organized according to the following description scheme (Figure 1) :

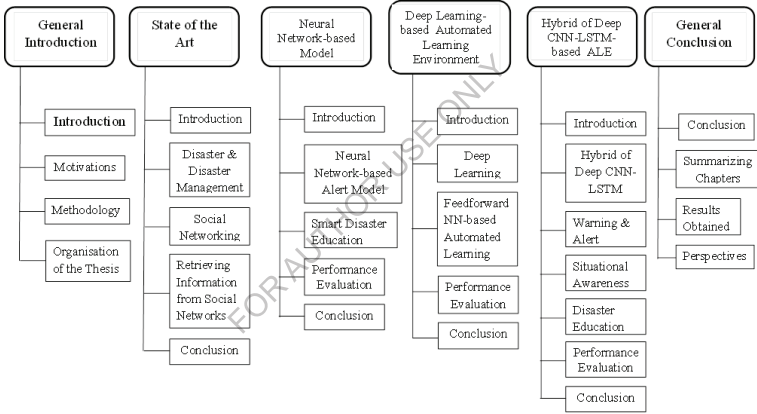


FIGURE 1 – Organization of the Thesis

The overview of disaster and disaster management models is presented in the Chapter 1. First of all, this chapter recalls the modelisation of disaster, after presenting all its definitions. This mathematical modelisation of disaster and disaster management allow the researchers to have an idea of different aspects and concepts necessary to save lives.

In Chapter 2, we present our basic implementation of the first alert model based on automated neural network for extracting relevant content from Twitter and Facebook using keywords and hashtags of disaster management. This Chapter allows us to begin designing et implementing the heart of the automated learning environment.

The chapter 3 represents the first enhancement of the automated learning environment. It is based on deep learning, namely Feedforward Neural Network and allows extracting, in real time, information on catastrophic events from multiple sources (all the Web) : it immediately alerts disaster managers to make quick and efficient decisions that could save lives. Indeed, we propose a new ad hoc real-time alert model for the management of disaster, whether natural or anthropogenic, based on a new multi-view recovery model from multiple sources. Such an approach is really useful for local disaster monitoring, but also for helping to make appropriate decisions.

Chapter 4 presents an other improvement of our real-time automated learning environment that is based on a Hybrid of Deep Convolutional LSTM neural network.

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"There is something fascinating about science. One gets such wholesale returns of conjecture out of such a trifling investment of fact."

Mark Twain

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Chapitre 1

State of the Art of relevant contents Retrieval from Social media

Summary

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1.1 Introduction

THIS work deals with the problem of extracting relevant information from social networks in crisis event. It begins by presenting conceptually disaster and disaster management, followed by the presentation of social networks and the different models used to extract the relevant information of a possible disaster.

Then, it presents a classification of all the mathematical models used for extracting relevant information, from SVM to neural learning. It consists of integrating artificial intelligence technologies, social media, disaster management and neural learning. It

consists of an automated learning environment based on an extension of the real-time alert model used for the management of natural and anthropogenic disasters that integrates encapsulations from multiple sources and retrieves information by combining multiple research results. This experience provides a backdrop for our automated learning environment and provides for it some great ideas.

Finally, this work ends by summarizing the main features of this approach, providing some references to related work, and discussing some issues related to artificial intelligence, social media and disaster management.

1.2 Disaster & Disaster Management

1.2.1 Disaster

Disasters such as earthquakes, flooding, fires, terrorist attacks and tsunamis result in disastrous human suffering, property destruction and other adverse effects. In addition to existing disasters, several anthropogenic disasters have arisen over the past two decades, primarily due to globalization, interconnected networks and substantial technological growth. Product forgery, biological risks, terrorism and ecological terrorism include anthropogenic disasters[10, 11]. The planet has undergone several significant natural and/or anthropogenic catastrophes of all time in recent years. Biological, geological, seismic, hydrological or natural processes such as cyclones, earthquakes, tsunamis, floods, forest fires, landslides, sandstorms and volcanic eruptions or hydro-meteorological paroxysms (*exceptional precipitation*), pandemics (*pandemic of the coronavirus such as Covid'19 and its variants*) [12] or human processes such as simple precipitation) are often modified by species. Indeed, the SARS-CoV-2 version (VUI 202012/01 renamed VOC 202012/01) has emerged rapidly from south-east England since September 2020, and many questions continue to arise. Dozens of other nations, including Europe¹, have reportedly identified the variant. Table 1.2 gives an overview of recent natural and anthropogenic disasters, and their damage.

From these cases, we find that we have 160,000 deaths and 60 million injured in 27 years (from 1980 to 2017), although we have serious damage only in the five years (from 2012 to 2017) with 32,454 dead, 3,355 injured, 6,639 missing, more than 83,000 hectares burned, 350 homes destroyed and significant damage. The urgency and significance of loss estimation and the need for decision support resources have been reaffirmed by recent threats from these disasters. In order to fulfill these needs and requirements, various models for disaster management[2, 6, 11, 13, 14, 15, 16, 17, 18] have been studied and built. The following are some of the major disaster management activities,

1. <https://www.cdc.gov/coronavirus/2019-ncov/transmission/variant.html>

TABLE 1.1 – Latest catastrophic events.

No	<i>Catastrophic event</i>	<i>Period</i>	<i>Damage</i>
1	Forest Fire Haiti	Oct 2007	230,000, 220,000
2	Earthquake California	Jan 2010	203 deaths, 6,152.9 Km ² ravaged lands
3	Floods Thailand	Jul 2011	815 Deaths
4	Tsunami earthquake Japan	Apr 2011	15,896 deaths, 6,157 injuries, 2,537 missing
5	Hurricane Sandy USA	Oct 2012	220 Deaths
6	Typhoo Haiyan Philippines	Nov 2013	26,626 injuries
7	Flood of the elbe Germany	Jun 2013	25 deaths
8	Subway bombing Russia	Apr 2017	15 deaths, 50 injuries
9	Suicide bombing England	May 2017	22 Deaths, 116 injuries
10	Three explosions Indonesia	May 2018	9 deaths, 40 injuries
11	Japan Floods Japan	Jun 2018	235 deaths, 13 missing
12	Indonesian earthquake	Sep 2018	2,000 deaths, 1.5 million injuries
13	Earthquake Fire Haiti	Oct 2018	18 Deaths, 548 injuries
14	Terrorist Attack Strasbourg	Dec 2018	5 deaths, 10 injuries
15	Kivu Ebola epidemic Congo	Aug 2018-Jun 2020	14,739,450 affected, 1,162 healed, 2,299 deaths
16	Coronavirus Pandemic COVID-19	Jan 21 st -Jul 23 rd , 2020	14,739,450 affected, 8,332,461 healed, 610,776 deaths

including hazard evaluation, risk management, mitigation, preparedness, response and recovery. When disaster strikes, people seek information and ways [10, 15, 16, 17] to provide data and assist those in need. Disasters inspire altruism, where individuals support those who are in distress or suffering from the disaster. Information on the protection of people and goods, as well as sources of aid, are among the most common forms of online assistance in the event of a disaster. As a consequence, study and experience in disaster management frequently refers to the formula 1.1 [11] of the following type :

$$\mathbf{R} = \mathbf{D} * (\mathbf{V} - \mathbf{Res}) \quad (1.1)$$

Where $\mathbf{R}$ is the risk that is the likelihood or expectation of a failure, $\mathbf{D}$ is a danger that is a risk-threatening situation. The vulnerability $\mathbf{V}$ is the extent to which individuals or objects are likely to be affected; and the tools $\mathbf{Res}$ are the assets in place that decrease the effects of hazards.

Disaster is defined [19] as a complex problem that must be addressed using a multidimensional and multiplatform framework to collect information. It is characterized as a severe disruption to the functioning of society [20] that involves extensive losses to humans, materials or the environment. Figure 1.1 shows the classification of various disasters. There are two main types of disasters : simple where the structure of the community remains intact and composed where the structure and function of the community are disrupted.

While disaster avoidance is not feasible, the likelihood of a disaster can be reduced by managing risk and vulnerability and enhancing individuals and communities' capacity to resolve these risks, in accordance with the previous formula (1). Disasters may differ in terms of types and impacts on human communities through knowing the source of common danger [11]. It is possible to classify disasters into two major categories : natural and anthropogenic. Natural disasters consist of disasters that are geophysical, biological and hydro-meteorological. Anthropogenic disasters consist of acts of terrorism, simple accidents and accidents involving technology. The threat, vulnerability and capability relationship is generally defined by the formula 1.2 [11] as follows :

$$\mathbf{Dis} = \mathbf{D} * (\mathbf{V} / \mathbf{A}) \quad (1.2)$$

Where **Dis** is the disaster, **D** is the danger to the disaster, **V** is the vulnerability to danger and **A** is the ability to overcome a disaster. This equation clearly shows that the risk of disaster for a community increases because of the high probability of a disaster (*danger*) and its inability to cope with the disaster. However, the use of adequate knowledge, skills and resources for disaster phases of preparedness and response can counteract the overall risk of a disaster.

Catastrophes are events that are fast-paced. Slow and chronic social disruptions [15], however, are important to theorize as catastrophes because they can have a greater effect than rapidly caused disasters.

Over the past two decades, many disasters have emerged in addition to existing disasters, mainly anthropogenic disasters due to globalization, interconnected networks and considerable technological development [2], [16], [17]. Anthropogenic disasters include product forgery, biological threats, terrorism and ecological terrorism. Figure 1 summarizes few examples of disasters with their type, location, year and number of victims.

Recent threats from these disasters have reaffirmed the urgency and importance of loss assessment and the need for decision support tools.

While it is not possible to prevent a disaster, the risk of a disaster can be minimized by controlling risk and vulnerability, and improving the capacity of individuals and communities to overcome these risks. Understanding the source of common danger, disasters can vary in terms of types and impacts on human societies [11], [17]. Disasters can be classified into two main categories : natural and anthropogenic. Natural disaster include geophysical, biological and hydro-meteorological disasters. As for anthropogenic disasters, they consist of terrorist acts, simple accidents and technological accidents. Figure 1.3 summarizes our new classification of disasters.

Disaster is said technological if it is due to a technical or human failure, occurred in a facility classified as dangerous for the protection of the environment, namely a

factory, a depot, a site, a quarry and damaged a large number of real estate, private or professional. Among these technological disasters, there are nuclear disasters and industrial disasters. Industrial disasters can be the result of negligence or incompetence, or any combination of these factors. These can usually result in more or fewer deaths, injuries or disappearances, as well as significant property damage. These consequences can be immediate or delayed. A nuclear disaster is a serious accidental industrial event, the consequences of which are related to the presence of radioactive materials. Figure 2 shows Global assessment of the Covid-19, as July 19th, 2020.

Terrorism is a crime against human life in violation of national and international laws while having a political purpose that would influence or change the policy of a country by terrorizing its civilian population.

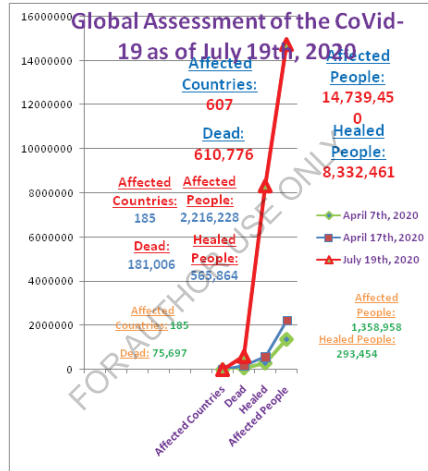


FIGURE 1.1 – Few examples of disasters with their type, location, year and number of victims.

Disasters are events that are fast-paced. Slow and chronic social disruptions[15], however, are important to theorize as disasters because they can have a greater effect than rapidly caused disasters.

When disaster strikes, people seek information and ways to provide information to help others. Disasters promote altruism, where people help those who are in difficulty or who suffer from the disaster. Among the most common types of online help in a disaster, they are in the form of information on the safety of people and goods which are more frequent and faster than that of official reports. This equation clearly shows that the risk of disaster for a community increases because of the high probability of a disaster (danger) and its inability to cope with the disaster. However, the use of

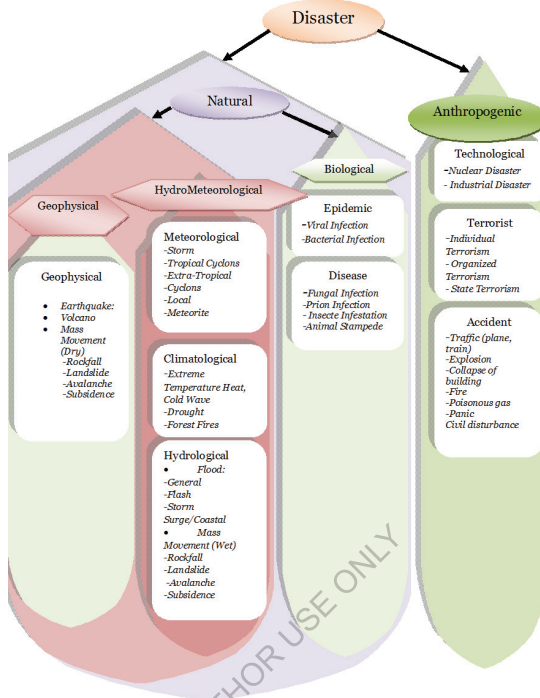


FIGURE 1.2 – Our new classification of disasters.

adequate knowledge, skills and resources for disaster preparedness and response can counteract the overall risk of a disaster.

1.2.2 Disaster management

The following stages are included in the disaster recovery cycle : warning, planning, action, prevention, mitigation and restoration. In disaster management, there are at least six key elements [21] : Prevention, Mitigation, Planning, Response and Relief, Restoration and Reconstruction. The process of disaster management, however, is defined in four phases, namely : [22] mitigation (before disaster), [21] preparedness (before disaster), [22] response (during disaster) and [21] recovery (after disaster).

The urgency and significance of loss estimation and the need for decision support resources have been reaffirmed by recent threats from these disasters. In order to fulfill these needs and requirements, various models for disaster management [2, 6, 11, 14, 15, 16, 17, 18] have been studied, designed and developed. The following are some of the major disaster management activities, including hazard evaluation, risk management,

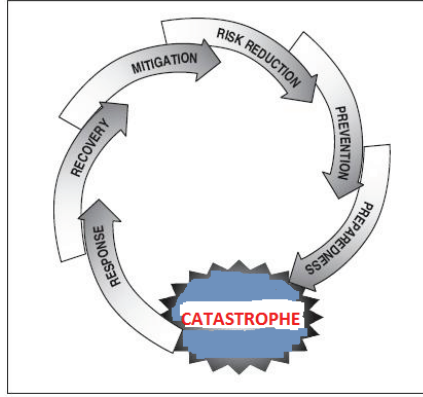


FIGURE 1.3 – Disaster Management Cycles

2

mitigation, preparedness, response and recovery.

In catastrophe management, there are at least six key elements [11], [17] : Prevention, Mitigation, Planning, Response (Relief), Restoration and Reconstruction. The emergency management process, however, is defined in four phases, namely [17] : mitigation (before disaster) [22], preparedness (before disaster) [20], response (during disaster) [22] and recovery (after disaster) [11], [20] (see Figure 1.4).

- Mitigation (before the disaster) : is the process of planning and taking long-term risk reduction measures for people and property in the event of a disaster [23]. This eliminates the risks before a disaster occurs.
- Preparedness (before the disaster) : is planning how to respond to a disaster [21].
- Response (in case of disaster) : consists mainly of conducting an assessment of the situation and implementing the strategies developed during the preparation phase [23]. It responds to the needs of the victims and take the necessary steps such as public warning, safe evacuation, management of persons or victims, measures to restore the basic structure of life and short-term plans to restore the normalcy of daily life [2].
- Recovery (after a disaster) : is about ensuring that all community activities and systems return to the normal situation [22]. These actions can be classified into two main categories, namely short-term recovery and recovery.

Disaster management is roughly categorized into two types [6, 17] namely structural measures (SM) and non-structural measures (NSM). SM can reduce physical damage directly related to disasters (eg. embankments, seismic buildings and evacuation sites).

However, they are expensive, time-consuming, and can not necessarily mitigate the damage caused by unexpected events. On the other hand, NSM are based on knowledge or practice; they can provide people with motivation and knowledge of disaster management at a relatively low cost (eg. policy and law development, evacuation drills and disaster education).

1.2.3 Disaster Management Models

In this conceptual framework and theoretical chaos, discrepancies and some variations between different models of disaster management have resulted in complications. While the scope of disaster management calls for templates to be used, [24]. Well-formed[25] typology can be very useful in maintaining discipline and eliminating complications in a chaotic environment. As for Disaster Management, the Classical Model, Computer Model and Disaster Management Social Networking Model are as follows :

1.2.3.1 Classical Disaster Management Model

By preventative measures, we can reduce the seismic risk, starting with the citizen's knowledge by teaching him the attitude to take before, during and after the earthquake, then reducing the seismic vulnerability of buildings, which can restrict the damage, without forgetting the cooperation between all the volunteers (solidarity action). Part of the popular wisdom of disaster management is that communication and collaboration be facilitated by personal familiarity, not just institutional contact.

Collaboration. For coping with natural and technological threats, disasters and the effects of terrorism, collaboration is an essential base. The disaster management field and profession is also developing into a more collaborative enterprise. The following equation models the collaboration :

$$\mathbf{DM} \rightarrow \mathbf{CE} = \mathbf{VC} - \mathbf{D} \quad (1.3)$$

with :

$$\mathbf{VC} = \{ VC_i \} \text{ with } i \in [1, N] \quad (1.4)$$

Where :

DM is the Disaster Management,

CE is the Collaborative Enterprise,

VC is the set of Volunteers Contents and

D is the set of Content of Volunteers in Double and

VC_i is the i^{st} Volunteer Content.

This evolution has increasingly shifted from top to bottom beyond the classical bureaucratic model to become a more complex and versatile [26] network model that promotes multi-organizational, intergovernmental and intersectoral cooperation. The partnership between all the volunteers as we see it is shown in figure 1.2.3.3.

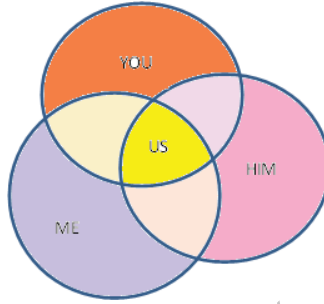


FIGURE 1.4 – Collaboration.

In disaster management, coordination is so critical, and command and control methods are so problematic that it becomes a monumental challenge to establish and maintain the required ties, and even a necessary task when dealing with disasters. This capacity is strengthened by regular engagement, including involvement in planning and training exercises. Collaboration has always been a skill for us [26] because of the use of volunteerism and community participation. Volunteers provide leading-edge capacity and links to community resources. Organizational and individual volunteer mobilization often serves a social and psychological purpose : it brings people together and gives them a sense of effectiveness. More adaptive leadership will enable organizational learning and make adaptation and improvisation easier.

Coordination. A multitude of social and behavioral research poses coordination as a significant obstacle for people, associations and organizations responding to disasters [27]. Furthermore, for emergency management preparation, coordination and knowledge sharing between various teams are prerequisites [27]. Channels of communication developed during the mitigation process serve as a basis [27, 28] for meaningful coordination and contribute to improving inter and intra-organizational resilience and cooperation and plays a big role in disaster risk reduction.

Dissemination of information. Transmit and/or exchange relevant information containing daily "updates" instead of a particular "warning" about a disaster, such as "weather updates" "traffic alerts" and "news". These types of data help [27] to keep people aware of their climate.

Issue warnings. In addition to daily 'information notifications' [27, 28], accurate and timely warnings play an important part in disaster management.

Communication. In all phases of disaster management, contact between community members remains important in terms of communication. They interact with each other during the mitigation process, either to keep in contact or to help each other planning : they must communicate during disasters (*or just after disasters*) to update their status and share about disaster ravages [15]. Moreover, there is also evidence that in the majority of disasters, the local communities and local authorities affected are the best ones to respond immediately. As a consequence, in emergency operations, the local group plays a critical role.

Moral / emotional support. One of the key roles of the media during and after a disastrous incident is to respond to the needs of people for emotional support, camaraderie and group ties. Individuals are actively seeking media emotional support [15], which provides isolated members of the group up to the time span identified by the impact of the event. Supportive action [15] is illustrated when people gather to express their gratitude to disaster responders and offer moral support to disaster victims.

1.2.3.2 Computer Model for Disaster Management

Damage evaluation is one of the main criteria of understanding of the situation in order to consider the nature of the devastation and also to prepare the relief accordingly. It is important to integrate humanitarian principles [29] into the design requirements of an information system. First of all, it must promote the production of disaster management skills with, for instance, disaster education or simulations.

Disaster Education Disaster education plays a critical role in motivating community members to improve disaster management skills. [16] In schools, industries and neighborhoods, evacuation exercises are also performed.

There are also GBED (*Games Based-Evacuation Drill : GBED*) [6] evacuation drills operating with motion hazard mapping (*Motion Hazard Map : MHM*) on a tablet with a GPS receiver and smart devices (*for example, tablets and smart glasses*). In disaster situations, virtual children have different reactions [30], while adults (*ie, HMD carriers*) are required to provide these virtual children with sufficient evacuation instructions while looking at situations of virtual disasters. Disaster Education Based Services [16, 31], as Paradigmatic Tourism : PT) that integrates Games Based-Evacuation Drill (GBED) and Black Tourism as a place-based disaster education., are other game-based evacuation exercises (*GBED*). Penumbral Tourism (*PT*) [32] is a place-based disaster education that uses disaster simulation in the real world.

A disaster fantasy game based on Tangible Bits [33] is developed. Tower defense game [9] improves ability to prepare for floods, to evacuate a three-dimensional vir-

tual (3D) world [34], immersive environments of virtual reality [35], Head-Mounted Displays (HMD) and other platforms [36]. Players can learn how to organize disaster response in Geo-fencing MRG[8]. As for GBED, when traveling to an evacuation location, participants often view digital documents on a portable computer, including electronic tablets and smart glasses.

Simulations Advanced models and broad data analyses have led to innovative disaster management methods being developed by visualizing disaster incidents that have not yet been realized.

Using Motion Danger Map (MHM) and smart devices, the tsunami evacuation drill (TED) framework [6] is built by simulation by configuring tsunami simulation using Google Maps. The causes of street flooding have been discovered by observations, road profiles and flood simulations [37] and suitable solutions have been suggested.

A multi-method simulation environment for humanitarian supply chains [38] is planned, prototyped and evaluated to create detailed models of relevant humanitarian supply chains to support decision-making process of humanitarian actors. Floods caused by short and abundant showers [37] have been analyzed and other variables have been defined using topography derived from light detection and telemetry (LiDAR), flood simulations, to find relatively inexpensive solutions to the traffic problems created by the floods.

1.2.3.3 Social Networking Model for Disaster Management

In general, the media plays a very significant role in disaster management. In the various stages of crisis management, the didactic role of the media differs only in content. Audiences seek information about risk, not preparedness, during the planning stage. During the impact process, during this scary moment, they get emotional support from the media, and connections to the outside world that break the isolation. The media focuses on the most affected areas after the disaster, providing estimates of harm and loss, and assisting communities in their recovery efforts. For recovery, after the impact, they want to know the conditions of the other communities. However, it is a difficult task to understand these voluminous and high velocity data, which requires the following crowdsourcing, crowdtasking and Collaborative Disaster Management.

Collaborative Disaster Management Large paper charts retain a distinct advantage in some situations, such as disaster response, in their combination of high resolution and portability : it is called geo-collaboration[39], a community work enabled by geo-spatial information technology on the problems of geographical scale.

To promote visualization, asynchronous and online interaction between actors, promoting distributed spatial and temporal cognition, a geo-collaborative, Web-enabled framework [40] is designed to target the unique characteristics of mobile and ubiquitous

computing environments.

Crowdsourcing and Crowdtasking in Disaster Management There is an evaluation of the advantages of work-sharing networks and social networking models (information collection, quasi-journalistic editing and crowdsourcing) in disaster management. Using motivational analysis to assess the most likely essential app features that will optimize continued user interaction, a modern method of developing a community-based computing environment acts as a real-time dashboard for government agencies responsible for monitoring populations during disasters. (See figure 1.6)

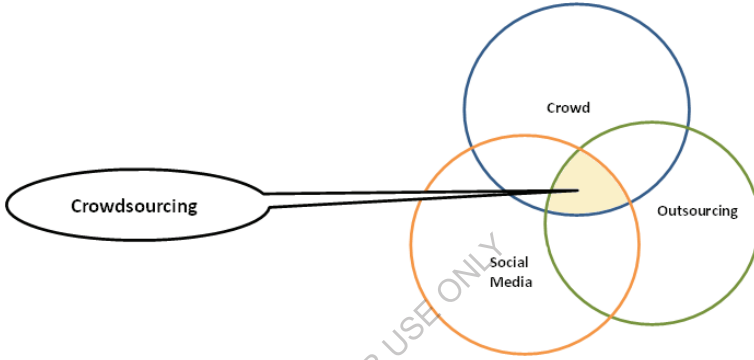


FIGURE 1.5 – Crowdsourcing.

The continued engagement of its users is measured by the performance of these community-based computer systems such as eBayanihan [41].

Crowdsourcing [41] can be a feasible production instrument that shows that extrinsic motives far outweigh extrinsic motivations (such as monetary reward).

In many occasions, the merits of unaffiliated volunteers have been shown. Tools such as Ushahidi[41] allow people to quickly access relevant information, such as reports on crisis situations and needs in their community, based on their geographical position.

To simulate involvement in crisis events, the motivation of volunteers in a serious gambling scenario is significant.

For crisis management, computerized application guidance systems[42], known as public safety systems, are used to rapidly organize public emergency services and save lives and property.

In recent years, a growing number of studies have examined the use of social networking data to gain knowledge of areas of human activity as diverse as the detection of disease as epidemics [43] and predicting the stock market. Social networking has emerged as a potential resource for improving natural or anthropogenic disasters [44].

To reduce their impact on humankind, the various management tasks during all phases of disasters [45] have growing information needs centered on the human being to facilitate better decision-making with a view to reducing loss of life and property. Social networks, which record a large amount of human activities, have attracted more and more attention from researchers [46]. Thus, their use, including micro-blogging such as Twitter, has spread, especially during disasters. Social networking is analyzed according to the four dimensions (*space, time, content and network*) [45] while summarizing the common techniques for extracting these dimensions and proposing some methods accordingly. While Dat Tien Nguyen & al. [47] introduce classification methods based on the neural network to identify tweets useful during a crisis situation.

Different social networks have different characteristics and are therefore more or less suitable for use during disasters depending on the phase and the type of disaster [20]. As Joao Porto de Albuquerque & al. [44] who present an approach to improve the identification of relevant social networking messages based on the relationships between geo-referenced social networking messages as voluntary geographic information and the geographical characteristics of data derived flood phenomena authoritative. As for the intensity of the earthquakes from information of the people who undergo them, the information [48] are compared with the 'reports' of markers and Tweets. Social networking such as wikis and collaborative workspaces are used to manage the 2010 earthquake in Haiti as the primary knowledge-sharing mechanisms [27, 49]. To locate survivors in a collapsed building and on fire, a new method [50] is proposed for indoor location in a building that collapses and burns with a disaster.

For coordinating relief activities from information from real-time Tweets, an alert system, which is based on Twitter, effectively managed complete disaster information because of its speed of communication, its reach and the quality of information [6]. The AIDR (*Artificial Intelligence for Disaster Response*) system combines human intelligence and artificial intelligence to categorize crisis-related messages in the event of sudden onset of disasters [3, 46]. Finally, for the exploration of social networking data, several techniques [51] could be described to make Twitter data more usable by emergency services and explore what information can be extracted from Tweets. A social networking image processing pipeline that combines human intelligence and artificial intelligence [52] captures and filters the content of social networking images and extract exploitable information during an ongoing crisis event through a hybrid crowdsourcing and machine learning approach. An end-to-end social networking image processing system called Image4Act [18] aims to collect and classify the content of images posted on social networking platforms, and to help humanitarian organizations to become aware of the situation and to launch relief operations. A comprehensive conceptual approach to geo-collaboration is applied to a range of prototype systems

that support both identical and different group activities [53]. Another generic coordination architecture targets, in particular, the specific characteristics of the mobile and ubiquitous computing environments that are required for collaborative applications [40]. While Brian M. Tomaszewski [54] & Alan M. MacEachren [53] developed a geo-collaborative, The work of Simone Wurster & al. [55] provides an approach for evaluating the benefits of work-sharing systems in disaster risk reduction. While Amisha M. Mehta & al. [56] have identified and evaluated three models of social networking use in disaster management, namely : information gathering, quasi-journalistic editing and crowdsourcing.

However, a new method of designing a community-based computing environment involves integrating information in the field into eBayanihan [19] which is Web-based for citizen reporting in addition to collecting information in social media. Michael Middelhoff & al. [41] aim to test the usability and acceptance of various methods and tools facilitating crisis communication via multiple channels. Jasmin Pielorz & al. [57] proposed the use of Open Street Map (*OSM*) driving distance instead of the previously used flight distance that better reflects the interests of volunteers and first responders. On the other hand, computerized application guidance systems, known as public safety systems, are used for disaster management to quickly coordinate emergency public services and save lives and property [42].

In post-disaster relief operations, it is assumed that all fixed cellular infrastructures do not be functional. The adaptability of P2P networks [58] must be exploited to respond to the characteristics of disaster situations. Peer-to-Peer networks have been used to interconnect field workers by just one active connection between a peer and the control room to perpetuate the disaster management system [59]. Due to the large autonomy of the peers involved leading to self-regulation behavior, such networks are very well prepared to address frequent changes in topology [60], information and positions. Rohit Sonawane & al. [61] used Wi-Fi peer communication in disaster management by developing an Android or iOS application that supports Peer-to-Peer Wi-Fi technology by enabling communication with others in disaster situations. With the increasing availability of mobile data communications, including Internet access in mobile networks, Peer-to-Peer applications from the wired network are also becoming available to mobile users [60].

M. Mecella & al. [62] developed a software infrastructure to support the collaborative work of human operators, in emergency / disaster scenarios, equipped with handheld devices (*PDA*s) and collaborating to achieve common goal : the whole team executes a process (*macro-processes*) and the different teams (*of different organizations*) collaborate by interleaving all the different processes.

M. Bortenschlager & al. [59] to voluntarily interconnect in the field with a single

active connection to the peer to peer sustain the claims management system. The data centers in the local network by the PAs, as to get from the geo-collaborative. They proposed to integrate Peer-to-Peer (*P2P*) concepts into Geo-Collaborative applications. This is a new prototype, a new concept is available in a new strategy to replication replication self-document to self to context.

Pradnya Mate & al. [63] developed a BitTorrent-like application for ad-hoc wireless networks on Android mobile phones. Except that, it is better to use small segments for downloading in order to have an increased parallelism with several peers involved. In addition, to maintain the quality of service of this application when transmitting data, they are considering future improvements.

A. Bhatnagar, et al. [64] implement a peer-to-peer protocol for priority message delivery to respond quickly to alert messages by creating a distributed phonebook for controlled message delivery and a response handling framework appropriate for alert messages. Certain restrictions remain necessary for networking, high dependence on the GSM operator, no possibility of dynamic role allocation, high performance allocation of a general-purpose mobile device, use of a method presumed based on the security policy of intensive messages.

R. Sonawane & al. [61] developed an Android or iOS application that supports Wi-Fi technology for free communication with others in the event of a disaster. This Wi-Fi Peer-to-Peer can be used in areas affected by a server, it allows users to search the network and available services before the connection. Unfortunately, this could cause interference during communication.

1.2.3.4 P2P Model for Disaster Management

Peer-to-peer (P2P) is a decentralized computer network model : the transactions that take place there take place between equally accountable nodes [58]. It is possible to exploit the adaptability of P2P networks [58] to meet the characteristics of disaster situations.

Figure 1.7 demonstrates the peer-to-peer network used to interconnect field staff to maintain and/or perpetuate the disaster management system [59] using a single active link between a peer and the control room. In these geo-collaborative implementations [59], P2P principles have thus been implemented.

In disaster management, there is also an Android or iOS application and an Android chat application [61] using Wi-Fi peer communication that supports Peer-to-Peer Wi-Fi technology by allowing communication in disaster situations with others.

The advantages of applying the P2P model [59] have been extended to the Disaster Management Geo-Collaboration as an alternative to GIS's most common client/server approaches and suggest the interconnection of P2P networks with mobile operators.

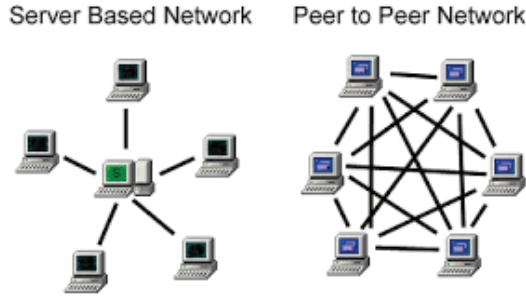


FIGURE 1.6 – Comparison between Server-based Network and Peer-to-Peer Network

Usually, wireless resources are scarce and differ in efficiency and availability, restricting the use in mobile environments of peer-to-peer applications.

Mobile P2P systems are a relatively new approach to integrating P2P principles with mobile computing technologies such as WirelessLAN, Bluetooth and telecommunication networks such as GSM, GPRS or UMTS [40], with over-frequent traditional changes in wireless settings that tend to be a challenge to conventional peer-to-peer systems. In the design of P2P systems, mobility support is becoming an increasingly important research subject.

To avoid this situation from lasting and growing, exchanging data on disaster situations should rapidly monitor the disaster. IoT/M2M network [65] is suggested to be paired with P2P cloud service to provide results as social services, such as SNS, for quick and smooth response in the event of a disaster with the eventual delay caused in the current commercial network. One active link between one peer and the control room at a time is sufficient to perpetuate the disaster management mechanism and increase the GeoCollaborative's availability.

P2P [66] algorithms for resource-limited and irregular (with no pre-existing infrastructure) WAHNs allow performance, scalability and fault tolerance to be achieved, where a spatially correlated group of nodes have crash simultaneously.

1.2.4 Discussion about Disaster Management Models

Different models for disaster recovery by academics and organizations have been suggested. Despite their success in some areas, disasters still pose a major challenge to sustainable growth. According to Nojavan et al. (2018) [67], which calls it *the strategic management*, the proposed typology in this study showed that the comprehensive model should include all three listed models due to the complementarity between disaster management models.

Qualitative vs Quantitative comparison. It may sound like the more accurate of the two is the quantitative method. But that's not the situation. Qualitative risk analysis is ideal for assessing probability and prioritizing risk in a simple way to understand, by rating severity in broader terms. It also makes it easier to recognise areas that need special attention, and can also be used to manage risk in real-time at any point of the project. That being said, there is an undeniably stronger combined approach³. Essentially, there are two sections of a single whole that allow you to assess *risk* comprehensively *risk level* of individual activities within the project schedule.

Disaster recovery consists of a layer of packages with subsystems. In order to provide scalable and extensible architectures, modular design unifies interfaces and data sharing. 3D visualization thus increases the interpretability of disaster data [68] and the efficacy of decision-making processes.

As for Disaster Education[6, 8, 9, 16, 30, 31, 33, 34, 35, 36, 69, 70, 71], Simulation[32] and Crowdsourcing[72] on Twitter only, there are several applications for Disaster Management in the Alert / Mitigation process. [14] on Twitter and Facebook for Forecasting and [15, 16, 17, 69] on all the Web and finally for Collaboration [26]. We also have in Preparedness phasis for Situational Awareness [69, 72, 73, 74, 75, 76] and for Damage Assessment [74, 75, 76, 77] on only Twitter and in Response phasis for Post-Disaster Coordination [68, 76]. But no application in Recovery phasis (see Table 1.3).

1.2.4.1 Complementarity between Disaster Management Models

Different models have been proposed for disaster management by researchers and agencies. Despite their efficiency in some locations, disasters are still a fundamental challenge in the way of sustainable development. Nojavan et al. (2018) [67] select the strategic management in the final phase as the global or overarching theme to integrate all the other themes. While the proposed typology in this study showed that the comprehensive model should include all the three mentioned elements because of the complementarity between disaster management models.

1.2.4.2 Quantitative/Qualitative Comparison between Disaster Management Models

Qualitative vs Quantitative comparison. Which is better for disaster management approaches Based on that description alone, it might sound like the quantitative approach is the more reliable of the two. But that is not the case. By ranking severity in broader terms, qualitative risk analysis is perfect for gauging probability and prio-

3. 09th Dec 19 Posted by Richard Wood Topics : Plan Risk Analysis

ritizing risk in a easy way to understand. It also makes it easier to identify areas that require special attention, and even, can be employed at any stage of the project to manage risk in realtime. That being said, a combined approach is unquestionably stronger. In essence, they are two parts of a single whole that enable you to comprehensively determine "risk level" of individual activities within the project schedule⁴.

1.2.4.3 Modular Architecture of Disaster Management

Disaster management system consists of a layer of subsystems packages. The modular design used in the Disaster management system unifies interfaces and data exchange to provide flexible and extensible architecture. For example, Subsequent 3D visualization improves the interpretability of disaster data and the effectiveness of decision-making processes [68].

1.2.4.4 Some Existing Disaster Management Software or Packages

Many applications exist on Disaster Management in Warning / Mitigation phasis as for Disaster Education[6, 8, 9, 30, 31, 33, 34, 35, 36, 69, 70, 71], Simulation [32] and Crowdsourcing [72] on only Twitter, for Forecasting [14, 78] on Twitter and Facebook and [15, 16] on all the Web and finally for Collaboration [26]. We also have in Preparedness phasis for Situational Awareness [72, 73, 74, 75, 76, 79, 80] and for Damage Assessment [74, 75, 76, 77, 80] on only Twitter and in Response phasis for Post-Disaster Coordination [68, 76]. But no application in Recovery phasis.

1.3 Social Networking

1.3.1 Introduction to Social Networks

Recent trends in the use of social networking highlight the fact that there is not only an increasing number of users of social networking applications, but also a significant increase in the number of such applications. In a short time social networks have invaded the daily lives of Internet users and Web professionals. The social media giants Facebook and Twitter were seen establishing, growing and evolving. They have been followed by a multitude of other more specific networks⁵ : Instagram, LinkedIn, etc The list is long. Social networks are essential reservoirs for the development of a disaster-stricken community, providing its members with a social history of past survival behaviors, social capital to cope, ways of assessing disaster risk and an effective means of transferring disaster-related information, especially with the arrival

4. 09th Dec 19 Posted by Richard Wood Topics : Schedule Risk Analysis

5. <https://www.blogdumoderateur.com/chiffres-reseaux-sociaux/>

of decentralized social networks (DOSN) with a high guarantee of confidentiality and security. Among the existing research studies, a group of studies identifies useful social networking information [4], using machine learning, to successfully extract structured information from unstructured textual Twitter messages. People use social networking to post situational updates in various forms [81] such as text messages, images and videos. Numerous studies [3, 82] have shown that this online information is useful for a quick response to disaster management. For example, after major disasters, researchers use Twitter data to know the number of people injured or dead [4], the most urgent needs affected people (*shelter, food, water, etc.*), donations, etc. Communication via social networking is direct, easy and instant and can simplify quick responses. Custom sites like Facebook, Twitter, Instagram, YouTube and Xing can subjectively offload the first contact [83] of authorities and service providers in crisis with strangers. These analyzes of the use of social networking in disasters have identified a distinct role for local users of the event (*or 'in the field'*), who are more likely to generate useful information to improve situational awareness [84]. Social networking can be considered as a practical and reliable emergency communication tool. While the predominant function of social networking remains social interaction, social networking sites are also considered the fourth most popular source of emergency information. The main objective of this research is to study the utility of social networking during disasters from the point of view of citizens. Different social networks have different characteristics and are therefore more or less suitable for use during disasters depending on the phase and the type of disaster. Social networking can support the exchange of information before, during and after a disaster in many ways. For example, disseminate timely alerts, publish regular updates on disaster and inform on available resources. The period before a disaster may also include activities such as gathering information from the public about potentially vulnerable regions or demographic groups. With the proliferation of social media, knowledge about disasters is transformed from expert knowledge to everyday knowledge co-produced by various stakeholders thanks to Web 2.0. The social networking can increase public awareness of different forms of contingency, placing it in the context of disaster knowledge, in light of the diverse needs of local communities.

In recent years, a growing number of studies have examined the use of social networking data to gain knowledge of areas of human activity as diverse as the detection of disease as epidemics and predicting the stock market. Social networking has emerged as a potential resource for improving natural or anthropogenic disasters. With regard to quantitative spatial and temporal analyzes, most of the existing work has sought to make sense of social networking data as an autonomous source by analyzing aggregated models. As Albuquerque (de) J. P. & al. [44] who present an approach to improve the identification of relevant social networking messages based on the relationships

TABLE 1.2 – Social media-based disaster management software or packages in different phases.

Disaster Management Actions	Disaster Management Phasis with Social Media Applications
1. Warning/Mitigation	
Disaster education	[6, 8, 9, 16, 17, 30, 31, 33, 34, 35, 36, 70, 71]
Simulation	Social Networking Model via a : only Twitter : [32] b : Twitter & Facebook : / c : All the Web : /
Forecasting	Social Networking Model via a : only Twitter : b : Twitter & Facebook : [14] c : All the Web : [15, 16]
Collaboration	Social Networking Model : [26]
Crowdsourcing	Social Networking Model via a : only Twitter : [72] b : Twitter & Facebook : / c : All the Web : /
2. Preparedness	
Risk assessment and reduction	Social Networking Model via a : only Twitter : / b : Twitter & Facebook : / c : All the Web : /
Situational Awareness	Social Networking Model via a : only Twitter : [72, 73, 74, 75, 76, 86] b : Twitter & Facebook : / c : All the Web : /
Damage Assessment	Social Networking Model via a : only Twitter : [74, 75, 76, 77, 86] b : Twitter & Facebook : / c : All the Web : /
3. Response	
Post-Disaster Coordination and Response	Social Networking Model via a : only Twitter : [68, 76] b : Twitter & Facebook : / c : All the Web : /
4. Recovery	
Normal Activities Resumption	Social Networking Model via a : only Twitter : / b : Twitter & Facebook : / c : All the Web : /

between geo-referenced social networking messages as voluntary geographic information, the geographical characteristics of data derived flood phenomena authoritative (*sensor data, hydrological data, and digital elevation models*) and very challenging biomedical data that is usually large, noised and imbalanced [85]. As for the intensity of the earthquakes from information of the people who undergo them, the information are compared with the 'reports' of markers and Tweets describing the same events to determine the intensity of the earthquakes and found a strong correlation between these two sources of information. Also, slip reports, produced by people filling out a Web form, arrive in a time frame similar to that of Tweets, although there are many more Tweets available. Social networking such as wikis and collaborative workspaces are used to manage the 2010 earthquake in Haiti as the primary knowledge-sharing mechanisms.

To locate survivors in a collapsed building and on fire, a new method is proposed for indoor location in a building that collapses and burns with a disaster by first estimating the two-dimensional attenuation distribution or three-dimensional in terms of the wireless signal used for localization, called the 'attenuation map', using wireless

tomography based on compressed detection. They finally locate a target node using the estimated attenuation map. For coordinating relief activities from information from real-time Tweets, an alert system, which is based on Twitter, effectively managed complete disaster information because of its speed of communication, its reach and the quality of information. The AIDR (*Artificial Intelligence for Disaster Response*) system combines human intelligence and artificial intelligence to categorize crisis-related messages in the event of sudden onset of disasters. This system, which exploits information from Tweets in real time from a disaster area to help coordinate relief activities, is now commonly used by the UN Office for the Coordination of Humanitarian Affairs (OCHA) and many other emergency services around the world [87].

Finally, for the exploration of social networking data, several techniques could be described to make Twitter data more usable by emergency services and explore what information can be extracted from Tweets. The images displayed on social networking platforms during disasters are analyzed to determine the level of damage caused by disasters. A social networking image processing pipeline that combines human intelligence and artificial intelligence [52] captures and filters the content of social networking images and extract exploitable information during an ongoing crisis event through a hybrid crowd-sourcing and machine learning approach. User-generated content on social networking in disasters is useful for disaster management and crisis response. On the other hand, an end-to-end social networking image processing system called Image4Act aims to collect and classify the content of images posted on social networking platforms, and to help humanitarian organizations to become aware of the situation and to launch relief operations. However, understanding these voluminous and high velocity data is a difficult task.

This section presents the most relevant related works, namely information retrieval models, in general, and those from several sources, in particular, and also alert models.

These are sites where people share feelings, whether they're Twitter, Facebook, Viber, Messenger, any forum on the Internet. The knowledge available on social networks varies from other Web sources (press articles, for example) in several respects. Such messages use less formal language, may include words from more than one language, may have different errors in grammar and spelling, and are, for the most part, unstructured, fuzzy and short-lived. Their length and content vary considerably [88]. We detected emotions using four types of features : interjections, blasphemy, emoticons and the general feeling of the message, which are widely used by individuals to convey emotions such as danger, surprise, happiness, etc. To identify these three types of features, we used a combination of par-of-speech (POS) tags, compiled lists of interjections and blasphemies on the Web for English and French and regular expression patterns for emoticons. From all online platforms automatically monitored by the Online Listening

Tool, namely Radian6 or one of its rivals [14, 15, 16, 17, 88, 89, 90, 91, 92] content can be gathered from websites to all social media. In fact, via Application Programming Interface (API) [16, 17, 88, 89, 91, 92] many networking platforms allow access to their data. The model, which fairly represents the essentials, is generated by online listening instruments, namely : harvesting contents, cleaning the data of non-informative information, enabling relevance thanks with the learning corpus obtained because of to the tagged messages, and analyzing the results.

1.3.2 Listening and Monitoring Online Social Media

TABLE 1.3 – Comparative table of all techniques and methods used in Models.

Ref	Identification Methods	Used OSN
[70]	Flood Disaster Game-based Learning	Twitter
[71]	Educational Purposes among the Faculty of Higher Education with Special Reference	Twitter
[74]	summarization with social-temporal context	Twitter
[75]	Capitalizing on a TREC Track to Build a Tweet Summarization Dataset	Twitter
[76]	semi-automated artificial intelligence-based classifier for Disaster Response	Twitter
[86]	Summarizing situational tweets in crisis scenarios : An extractive-abstractive approach	Twitter
[14]	Based on Artificial Neural Network (ANN)	Twitter & Facebook
[15]	Based on Artificial Neural Network (ANN)	All the Web
[16, 17]	Based on FeedForward Neural Network (FFNN)	All the Web
[93]	Based on Recurrent neural network (RNN) trained with Long Short-Term Memory (LSTM)	All the Web

Contents were collected from all online channels tracked automatically by the Online Listening Tool, namely Radian6 [92] or any its competitor, such as Awario, Brand24.com, Brandwatch, Mention, Keyhole, Socialert.net, SocialPilot.co, Simplify360, etc. from websites to social media, such as Twitter, Facebook, LinkedIn, Instagram, Google+, Youtube and so on. Actually, many networking platforms allow access to their data via Application Programming Interface (API) [88]. Online listening tools provide the model, which reasonably represents the essentials, namely : *harvesting contents* (such as conversations at the social media, news or any information in the Web), *cleaning the data* of duplication and replication content, *enabling relevance* (thanks to the neural learning, obtaining of relevant information with the learning corpus obtained thanks to the tagged messages) and *analyzing the results* (verification and analysis of the results).

Social networking forms an important part of online activities of Web users. There are two types of social networks : Centralized social networks and Decentralized social

networks. Current Online social networks (OSN) are Web services run on logically centralized infrastructure [94]. They use content distribution networks and thus distribute some of the load by caching for performance reasons, while keeping a central repository for user and application data. This centralized nature of Online Social Networks has several drawbacks including scalability, privacy, dependence on a provider, and need for being online for every transaction [95]. Web sites such as Facebook, MySpace and Orkut have millions of users using them every day. A decentralized online social network (DOSN) is a distributed system for social networking with no or limited dependency on any dedicated central infrastructure [94], while being a solution to the violation of privacy, especially thanks to P2P architecture [96].

Recent trends in the use of social networking highlight the fact that there is not only an increasing number of users of social networking applications, but also a significant increase in the number of such applications. In a short time social networks have invaded the daily lives of Internet users and Web professionals. The social media giants Facebook and Twitter were seen establishing, growing and evolving. They have been followed by a multitude of other more specific networks : Instagram, LinkedIn, etc. People use social networking to post situational updates in various forms [81] such as text messages, images and videos. Numerous studies [3, 15, 16, 17, 97] have shown that this online information is useful for a quick response to a particular situation. Communication via social networking is direct, easy and instant and can simplify quick responses. Custom sites like Facebook, Twitter, Instagram, YouTube and Xing can subjectively offload the first contact of authorities and service providers. These analyzes of the use of social networking in events have identified a distinct role for users, who are more likely to generate useful information to improve situational awareness [97]. Social networking can be considered as a practical and reliable emergency communication tool. While the predominant function of social networking remains social interaction, these sites are also considered the fourth most popular source of information. Different social networks have different characteristics and are therefore more or less suitable for use during a given situation. Social networking can support the exchange of information before, during and after an eventual event. With the proliferation of social media, knowledge is transformed from expert knowledge to everyday knowledge co-produced by various stakeholders thank to Web 2.0 [16, 17]. In recent years, a growing number of studies have examined the use of social networking data to gain knowledge of areas of human activity as diverse as the detection of disease as epidemics and predicting the stock market. However, understanding these voluminous and high velocity data is a difficult task. Contents were collected from all online channels tracked automatically by the Online Listening Tool, namely Radian6 [91, 92] or any its competitor, such as Awario, Brand24.com, Brandwatch, Mention, Keyhole, Socialert.net, SocialPilot.co,

Simplify360, etc. from websites to social media, such as Twitter, Facebook, LinkedIn, Instagram, Google+, Youtube and so on. Actually, many networking platforms allow access to their data via Application Programming Interface (API) [88]. Online listening tools provide the model, which reasonably represents the essentials, namely, as shown in figure 8 :

- 1 - Harvesting contents : such as conversations at the social media, news or any information in the Web ;
- 2 - Cleaning the data of duplication and replication content ; eliminating, from the content, any dubbed information like retweet, and any information harmful or redundant ;
- 3 - Enabling relevance : thanks to the neural learning, obtaining of relevant information by using the machine learning with the learning corpus obtained thanks to the tagged messages. These tagged messages are realized by volunteers ;
- 4 - and Analyzing the results : During this stage, the verification and analysis of the results is carried out in order to ensure adequacy so that it is obtained to build disaster information such as situational awareness, damage assessment and/or disaster education.

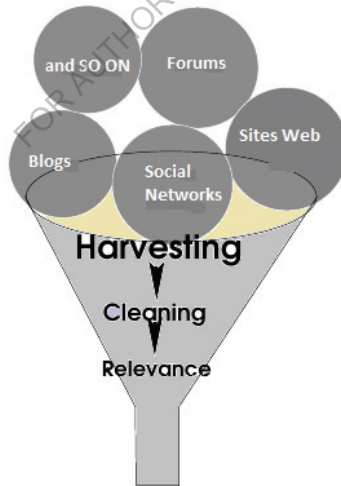


FIGURE 1.7 – Online Methodology Reflecting our Listening and Monitoring Approach.

Some of the benefits of using the Twitter and Facebook APIs include :

- 1 - using the development space of the two social media ;

- 2 - encouraging development environments ;
- 3 - scientific recognition of development environments ;
- 4 - encouraging other social media to involve themselves in research development ;
- 5 - helping social media to feel imbued with this desire for development and scientific research in parallel with their commercial activity ;
- 6 - and contributing to development of further improvements in networking services.

1.4 Information Retrieval Models from Social Networks

In a bilingual recovery, comparable document-aligned data [98] is used to easily learn patterns of common topics or word concepts. However, the preparation of comparable documented data requires a lot of human involvement and time. Therefore, such approaches are not appropriate when information about an ongoing event should be retrieved quickly. The work of Joseph A. Shaw and Edward A. Fox [99] focuses on methods for recovering separately from different sources and then combining them using data fusion techniques. Multi-view recovery based on deep learning that attempts to learn document embedding in a common space, where differences between different data sources would not exist. Unfortunately, this approach has low performance scores for all the methods used : indicating that the problem is difficult and requires better methods. According to Cynthia D. Balana [100], Google Crisis Response's (GCR) National Risk Management and Disaster Response Board (NDRRMC) had integrated Facebook and Twitter into its surveillance system and website for a quick response.

Whether it is Twitter, Facebook, Viber, Messenger, any forum or any thing in the Web, these are platforms where people often express emotions. Emotions are detected using four types of features, namely : interjections, blasphemy, emoticons, and the general feeling of the message.

Interjections, blasphemies and emoticons are widely used by individuals to convey emotions such as anger, surprise, happiness, etc. To identify these three types of features, a combination of POS tags is used in the English tagger (which contains tags for interjections, emoticons, etc.), compiled lists of interjections and blasphemies on the Web for English and regular expression patterns for emoticons.

1.4.1 Relevant Information Retrieval from Social Media

Most event detection methods are based on keywords (hashtags) used in tweets during catastrophic events to classify messages as real-time event reports, using a support vector machine (SVM). Tweets produced during the flood of the Elbe in June 2013 in Germany [44, 101] are examined, using only, statistical analysis to identify general spatial patterns in the occurrence of flood-related tweets that may be associated with the proximity and severity of floods. Comparable document-aligned data [98] are used to easily learn patterns of common topics or word concepts, in a bilingual recovery. However, the preparation of comparable documented data requires a lot of human involvement and time. Therefore, such approaches are not appropriate when information about an ongoing event must be retrieved quickly. Distributed situation awareness reports based on Twitter [72] activity during natural disasters are captured. We will study all artificial learning methods, from machine learning to deep learning, after an overview of techniques for retrieving relevant knowledge on social networks.

1.4.1.1 Classification Models

In Automated Learning, we distinguish regression problems from classification problems. Thus, we consider that the problems of predicting a quantitative variable are problems of regression while the problems of predicting a qualitative variable are problems of classification. Certain methods, such as logistic regression, are both regression methods in the sense of predicting how to belong to each of the classes and classification methods.

The performance of this system strongly depends on the selection of the appropriate model. We are now trying to select the most appropriate model for our proposed feature set based on specific criteria. Our learning approach is based on four advanced classification models for the feature set above, namely : We will study all the techniques for retrieving relevant knowledge on social networks, from Support Vector Classification to Neural Learning, including the Random Forest Classification.

1. Support Vector Machine : To solve regression problems, the approach used for support vector classification can also be expanded. Training points beyond a certain boundary are not taken into account in the cost function for building the support vector classification model. Therefore building a support vector classification model only depends on a subset of training data [102].
2. Random Forest : In order to control over-fitting and to increase predictive precision, Random forest generates a lot of decision tree based on random collection of data and variables and takes the notion of averaging. Every tree in the lot is developed from the training set using bootstrap sampling. In addition, when a

node is split during tree creation, the selected split is not the best split between all the features, but it is the best split between a random subset of features. The bias of the forest usually increases because of this, but also decreases due to techniques such as averaging its variance, which compensates for more than an increase in bias [103]. It is part of machine learning techniques, used in data mining and machine learning, and refer to a method based on a decision tree as a predictive model. Decision Tree Forests combine the concepts of random subspaces and bagging. Their algorithm performs learning on multiple decision trees driven on slightly different subsets of data.

3. Logistic Regression : It is a binomial regression model that best models a simple mathematical model with many real observations. Used in machine learning, it is a special case of generalized linear model. Logistic regression or logit model is a binomial regression model. As with all binomial regression models, it is best to model a simple mathematical model with many real observations. In other words, to associate with a vector of random variables $(x_1, \dots, x_K)$ a binomial random variable generically noted y . Logistic regression is a special case of a generalized linear model. It is widely used in machine learning.
4. Symbolic Regression : It consists in obtaining mathematical functions using a finite sampling of the values of independent variables and the associated values of dependent variables. Thanks to the artificial bee colony programming algorithm (ABCP) and recently the automatic multi-tree artificial bee colony programming algorithm (MTABCP), which remains on the association of the ABCP programming method and least squares, this method is used in Machine Learning. Symbolic regression constructs mathematical equations by composing both parameters and equational forms. In other words, it attempts to derive non-linear equations by simultaneously manipulating equational shapes and parameters while solving a given modeling problem. The symbolic regression method aims to find the best combination of variables (inputs and outputs), symbols and coefficients in order to develop an optimal model satisfying a set of fitness cases. The problem of symbolic regression can be considered as an optimization problem in order to find the best function of the variables, symbols and coefficients satisfying a performance criterion.
5. Naive Bayes : The Bayesian naive classification implements a naive Bayesian classifier belonging to the family of linear classifiers. A more appropriate term for the underlying probabilistic model could be *statistically independent model*. Depending on the nature of each probabilistic model, naive Bayesian classifiers can be effectively trained in a supervised learning context. Despite their *naive* design model and extremely simplistic basic assumptions, naive Bayesian clas-

sifiers have been more than efficient in many complex real world situations. In 2004, an article showed that there are theoretical reasons behind this unexpected effectiveness. However, another 2006 study shows that more recent approaches (reinforced trees, random forests) provide better results. The advantage of the naive Bayesian classifier is that it requires relatively little training data to estimate the parameters necessary for classification, namely averages and variances of the different variables.

6. Rule Based Classifier : The rule-based classification term can be used to refer to any classification system using IF-THEN rules for class prediction. Classification schemes published on rules generally include the following components :

- Rule induction algorithm consisting of the process of extracting the relevant IF-THEN rules from the data ;
- Rule classification measures : referring to certain values used to measure the usefulness of a rule to provide an accurate prediction. Rule ranking measures are often used in the rule induction algorithm to eliminate unnecessary rules and improve efficiency. They are also used in the class prediction algorithm to rank the rules which will then be used to predict the class of new cases.

We follow a simple approach : if any of the features mentioned above are present in content, we mark it as relevant. Table 1.4 presents the set of features for the content extraction task.

We follow a simple approach - if any of the features mentioned above are present in a content, it is marked as relevant.

TABLE 1.4 – The set of features for the content extraction task.

No	Descriptions
1	<i>Slang expressions. Example : Bail, Feeling blue, Buck, By the skin of (my / your / his / her) teeth, Creep (n.), Couch Potato, Cram, Crash, Down to earth, Drive up the wall, For Real, Going Dutch, The cold shoulder, ...</i>
2	<i>Word contractions formed by just the initials such as ' u ' for ' you '.</i>
3	<i>Word contractions with just a part or a syllable of the word, such as ' HAV '.</i>
4	<i>Numbers to replace words or syllables of the word as ' 2hav '.</i>
5	<i>Sentences formed by the initials of words.</i>
6	<i>Contracted forms of certain English words. Example : I' m, I' ve, I don ' t, I ' d.</i>
7	<i>Idioms or proverbs to convey certain ideas, principles and values that underlie English culture. Some combinations of words have a figurative meaning and essentially work with images. Example (A hot potato), (A penny for your thoughts), (Add insult to injury), (At the drop of a hat), (Back to the drawing board), (Ball is in your court).</i>
8	<i>Stenographic words consisting of initials, such as (u) for (you).</i>
9	<i>Onomatopoeia that reproducing or mimicking the sound of words or phrases to convey certain ideas, good or bad.</i>

1.4.2 Neural Learning

Neural Learning (NL) is an artificial intelligence technology that enables computers to learn without having been explicitly programmed to do so. To learn and increase, however, computers need data to analyze and train on [104]. Abiodun et al. (2018) [104] recommend that future research can focus on combining, into one network-wide application, various Automated Neural Networks (ANN) models, namely Machine Learning and Deep Learning.

In Automated Learning, we distinguish regression problems from classification problems. Thus, we consider that the problems of predicting a quantitative variable are problems of regression while the problems of predicting a qualitative variable are problems of classification. Certain methods, such as logistic regression, are both regression methods in the sense of predicting how to belong to each of the classes and classification methods.

Among the existing research studies, a group of studies identifies useful social networking information, using machine learning, to successfully extract structured information from unstructured textual social media contents.

Despite the fact that machine learning is not new concept, many people are still uncertain what it entails. It is a modern science that uses statistics, data mining, pattern recognition, and predictive analysis to identify patterns and make data predictions. At the end of the 1950s, the first algorithms were developed. The best known of these is none other than the Perceptron.

The training data is created from the June 2018 Japan Floods and the October 2010 Haiti earthquake disasters. This information, easily obtained using the neural network, is manually annotated by volunteers.

1.5 Conclusion

With defining and modeling concepts of disasters and disaster management, with proposing disaster classification, exploring and analyzing social media-based disaster management packages in different phases, this work aims to explore the potential of social networking in managing natural or anthropogenic disasters and shows the impact of social networking paradigm on the improvement of the disaster management process where interactions involving communities are discussed. They have their specific functional needs to act during the different phases of the disaster management process. In addition, the role of the communication means in the attenuation, response and recovery phases is also presented. We have explored the potential of P2P networks in managing natural or anthropogenic disasters. The adaptability of P2P networks [58] should be exploited to respond to the characteristics of disaster situations. Al-

though the importance of social networking has already been modeled and reported above, it is also important to empirically validate the potential of social networking in disasters through semi-structured interviews that will be conducted with disaster managers of social media. The choice of these managers will aim to obtain a good combination of cases based on their expertise and areas of operations. The objectives of these interviews are :

1. Explore the functional needs of disaster managers for action during the mitigation, response and recovery phases.
2. Identify the advantages and disadvantages of using social networking by disaster managers to act during mitigation, response and recovery phases.

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"There is something fascinating about science. One gets such wholesale returns of conjecture out of such a trifling investment of fact."

Mark Twain

FOR AUTHOR USE ONLY

Chapitre 2

Neural Networks-based Automated Learning Environment for Relevant Information Retrieval from Social media

Summary

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2.1 Introduction

IN recent years, various catastrophic events, summarized in Table 1.1, have occurred. With all these disasters, social media has enabled the affected population to quickly publish a considerable amount of disaster information to help decision makers make quick and good decisions [44, 80, 105, 106].

With the ability to share a message with a potentially important audience, social networks such as Facebook or Twitter, among others, are used to collect and disseminate relevant and up-to-date information [82], [105]. However, the quantity and quality of information provided during these critical periods and crisis situations can make it difficult to find useful and usable information. Indeed, social networks also serve to sabotage reports such as the Asiana flight accident in July 2013, a photo that

is published on Twitter 30 seconds after the accident [3], [107].

Social networks have become a potential resource for communication, detection, monitoring and disaster information extraction to improve the management of natural and anthropogenic disasters [3, 82, 105, 108, 109].

The use of social media to communicate timely information during crisis situations has become a common practice in recent years [109]. Major crises can generate millions of messages on social networks (*eg. 3.5 million messages in one day, during Hurricane Sandy in 2012, and more than 5,500 tweets published every second in the wake of the tsunami and the earthquake in Japan in 2011*) [110]. Manually checking each message and filtering out the relevant ones is a daunting task, if not impossible.

The link between social networks and disaster awareness has attracted increasing attention from researchers with different types of games and simulations for disaster education [3, 83, 98, 111]. These numerous studies have been carried out in recent years for the automatic identification of different types of content on social media [83, 98, 111].

With the proliferation of social networks, an ongoing event is under discussion on all these channels. Affected populations publish useful and timely information of various types, including reports of injuries or deaths, damage to infrastructure or urgent needs on different social networks. This important information obtained in the first hours can be of great help and can also reduce both human losses and economic damage [52, 112]. To get a complete view of the event, it is important to retrieve information from multiple sources. There are generally qualitative differences in the information obtained from different sources. However, when it comes to retrieving information posted on social networking sites and especially to alert people to make quick decisions, the task of disaster management is heavy.

The unambiguous nature of Twitter has enabled stakeholders to deliver relevant messages for the crisis and to access vast amounts of information they may not have. However, the Web has several sources of information on which an ongoing event is discussed. Several automated systems have been designed to help disaster managers identify and filter useful information posted on social networking sites [3]. Most of the work has focused on the use of social networks as a source of information on only a few phases of disaster management, including disaster response. However, few are devoted to warning and all methods deal with the retrieval of information from social networks. Generally, Twitter is the main source of information for these systems. The design of disaster management systems with different and diverse sources of information, especially social networks, is a real challenge.

In this research work, we propose a new real-time alert model for disaster management based on a neural network that integrates encapsulations from multiple sources,

and thus retrieves information by combining multiple search results. Thus, we provide, not only, a solution to this challenge, but also, we propose better performances in term of recovery than other similar solutions.

In our proposed solution [15], that is the enhancement of our previous work [14], we try to identify relevant content dealing with impending catastrophic events or just bursting. These disasters can be natural or anthropogenic. Once this information is retrieved (*distinguished, of course, from both large amounts of other content and also abusive information*), it can be used to alert disaster managers to make quick and effective decisions that could help people in need or to save lives. Our main contributions are listed below.

1. We develop a neural-based model that uses low-level content learning capabilities to automatically separate relevant information from redundant (*eg. retweets for Twitter*) or abusive (*eg. community-based*).
2. We develop a neural network-based model that uses content learning capabilities of multiple sources (*ie Twitter and Facebook*) to automatically retrieve relevant information using a set of keywords and hashtags respectively related to various catastrophic events whether natural or anthropogenic.
3. Keeping in mind the limitations of the previous work [14], we develop an event-independent alert model that can be used directly to filter content on multiple source at a time in future events. Experiments on multi-disaster-related tweet flows with diverse characteristics show that our proposed model outperforms vocabulary-based approaches. Our approach filters the content (*Twitter tweets and Facebook messages at a time*).
4. Once we have developed this real-time, neural-based alert model, using multi-source content learning capabilities (*ie Twitter and Facebook*) to automatically retrieve relevant information by using a set of keywords and hashtags respectively, relating to various catastrophic events whether natural or anthropogenic, we tested the proposed model on the two catastrophic events of this end of summer, namely, Japan Floods of June 28 to 09 July 2018 with 2000 dead, 2500 wounded and 330,000 homeless and earthquake of Haiti from 06 October 2018 with 18 dead and 548 wounded.
5. Having identified the relevant information, we have proposed an alert network propagation scheme that permits disaster managers, so they can quickly make effective decisions that would save human lives.

2.2 New Alert Model to Disaster Management

Our approach is most closely related to that of Koustav Rudra & al. [80], who studied Community Tweets, that is, abusive messages targeting specific religious / racial groups displayed during natural disasters. Except that they proposed methods to counter them by using anti-communal tweets published by some users during catastrophic events.

2.2.1 Information retrieval models from multiple sources

Vulic & al. [98] use comparable document-aligned data to easily learn patterns of common topics or word concepts, in a bilingual recovery. However, the preparation of comparable documented data requires a lot of human involvement and time. Therefore, such approaches are not appropriate when information about an ongoing event should be retrieved quickly. The work of Joseph A. Shaw and Edward A. Fox [99] focuses on methods for recovering separately from different sources and then combining them using data fusion techniques.

Multi-view recovery based on deep learning that attempts to learn document embedding in a common space, where differences between different data sources would not exist [78]. Unfortunately, this approach has low performance scores for all the methods used : indicating that the problem is difficult and requires better methods.

According to Cynthia D. Balana [100], Google Crisis Response's (GCR) National Risk Management and Disaster Response Board (NDRRMC) had integrated Facebook and Twitter into its surveillance system and website for a quick response.

TABLE 2.1 – Comparative table of all techniques and methods used in Alert Models.

Approach	Methods	Identification	OSN	Analysis
[4]	on keywords	Classify messages	Twitter	Real-time
[44]	Statistical analysis	Identifying spatial patterns	Twitter	/
[80]	Annotated data	Identify abusive contents	Twitter	/
[81]	Annotated data	/	Twitter	Real-time
[107]	/	Learning	Twitter	/
[108]	/	Learning	Twitter	/
[109]	Annotated data	/	Twitter	/
[115]	Annotated data	Notification system alerts earthquake	Twitter	/
[118]	Annotated data	CrowdsourcingMachine learning	Twitter	Real-time
[119]	Predefined geographic displays	/	Twitter	Real-time
[120]	on keywords	/	Twitter	/
Our Approach	Ap-keywordsAnnotated messages	Learning with NN	all the Web	Real-time analysis

TABLE 2.2 – Comparative table of all techniques and methods used in Models for retrieving information from multiple sources including our approach.

Approach	Exigences	Real-Time	Multiple Sources	Method
[78]	Low performance scores	/	/	Multi-view recovery based on deep learning
[98]	Requires a lot of human involvement and time	Not appropriate for real-time approach	/	Learning patterns with comparable documents
[99]	/	/	Recovering separately from different sources	Combining data fusion techniques
[100]	/	/	Facebook/Twitter	/
Our approach	Annotating manually information	Appropriate for real-time	Multiple sources (Facebook Twitter)	Learning with Neural Network

In this paper, we propose a new alert model for disaster management based on social networks, as [127] and [128]. Our proposed model is based on a semi-supervised inductive technique to use unlabeled, multi-source data, which is often abundant during a crisis event, with less data previously labeled than previous events (*Table 1.2, Table 1.3 and Table 1.4*).

2.2.1.1 Foundation of neural learning

Figure 1 shows the flow chart showing the functioning, during the first passages, of this alert model to learn the first information that will be manually annotated by volunteers, as shown by Algorithm 1.

Algorithm 1 : Learning information using annotated manually contents

```

1 : begin
2 :   input (a content  $V(w_1, w_2, ..., w_i, ..., w_n)$ )
3 :   while (  $w_i \in \{Keywords\}$  ) do
4 :     {information}  $\leftarrow w_i$ 
5 :     skip to the following content  $w_{i+1}$ 
6 :   endwhile
7 :   output listof information  $V(w_1, w_2, ..., w_i, ..., w_n)$ 
8 : end

```

This annotated information is integrated in the model to help the neural network in learning and thus retrieves relevant information that is used in alerting public opinion

TABLE 2.3 – Comparative table of all techniques and methods used in Models for retrieving information from multiple sources.

Approach	Exigences	Real-Time	Multiple Sources	Method
[78]	Low performance score	/	/	Multi-view recovery based on deep learning
[98]	Requires a lot of human involvement and time	Not appropriate for real-time approach	/	Learning patterns with comparable documents
[99]	/	/	Recovering separately from different sources	Combining using data fusion techniques
[100]	/	/	Facebook / Twitter	/
Our Approach	Annotating manually information	Appropriate for real-time	Multiple sources (Facebook Twitter)	Learning with Neural Network

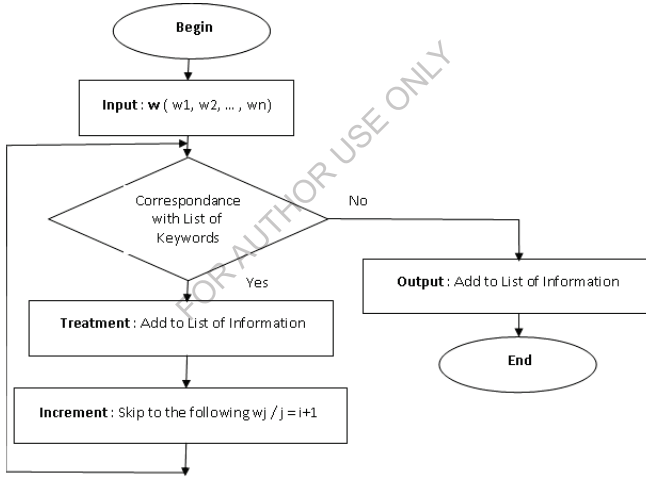


FIGURE 2.1 – Flow chart for determining, using a set of keywords, information that will be annotated manually to enrich the neural network in its possible learning.

and disaster managers to make appropriate decisions. Figure 1.2 shows how this alert model works.

Algorithm 2 : Determining relevant information with annotated information.

1 : begin

```

2:  input (a content  $V(w_1, w_2, \dots, w_i, \dots, w_n)$ )
3:  while ( $w_i \in \{\text{Keywords}\}$ ) do
4:       $\{\text{information}\} \leftarrow w_i$ 
5:      skip to the following content  $w_{i+1}$ 
6:  endwhile
7:  output list of information to be annotated
8:  while ( $w_i \in \{\text{information annotated manually}\}$ ) do
9:       $\{\text{information}\} \leftarrow w_i$ 
10:     skip to the following content  $w_{i+1}$ 
11:  endwhile
12:  output list of information  $V(w_1, w_2, \dots, w_i, \dots, w_n)$ 
13: end

```

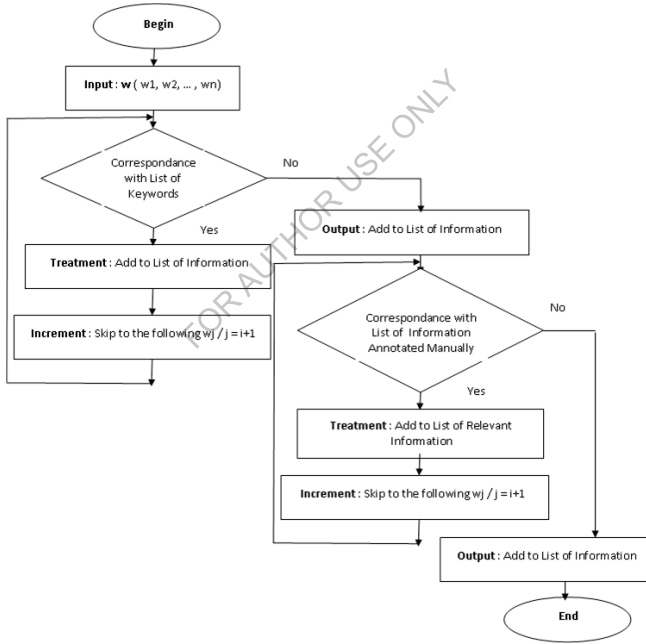


FIGURE 2.2 – Flow chart for determining relevant information using a set of manually annotated information

Figure 2.2 shows the flow chart showing the functioning, during the following passages, of this alert model to learn the relevant information that will be used to alert

public opinion and in particular disaster managers so they can take quick and effective decisions that can save lives, as shown in Algorithm 2.

We use a neural network with a single hidden layer that takes e_i as input and gives e_k as output.

The input to the network is a content :

$$t = (w_1, ..., w_i, ..., w_n)$$

containing words each coming from a finite vocabulary.

Let :

$$e_i \in C^n, \text{ with } i \in [1, N] \text{ and } e_i = (w_{i1}, w_{i2}, ..., w_{in})$$

containing words each coming from each a finite vocabulary V , the incorporation of a content of the source message i relevant for, at least, a keyword :

$$H_j \in \mathbf{H} \quad \text{with } j \in [1, M].$$

We want the learning of a generic space with the feed-forward neural network, as :

$$\mathbf{E} = \{e_k; \quad k \in [1, K]\}$$

which normalizes the differences:

$$E = [\mathbf{E} - \mathbf{RDF}]$$

where :

$$\mathbf{RDF} = [\mathbf{R} + \mathbf{D} + \mathbf{F}]$$

Thanks to the neural network, the transformation of e_i into e_k can be explained by :

$$e_i \rightarrow e_k = \{e_i; \quad \text{such as } e_i \text{ is relevant for } H_j \text{ and } w_i;$$

$$H_j \in \mathbf{H} \quad \text{with } j \in [1, M]\}$$

where

$$w_l \in \mathbf{W}; \quad \text{with } l \in [1, L]; \quad e_i \in [\mathbf{R} + \mathbf{D} + \mathbf{F}];$$

with $k \in [1, K]; \quad i \in [1, N]$

D, **R** and **F** represent respectively the set of duplicate retweets, duplicate contents and false alerts. The objective is then to maximize the size K of the set **E**. Figure 1.2 shows the Artificial Neural Network Architecture.

De Albuquerque & al. [52] used only statistical analysis to identify general spatial patterns in the occurrence of flood-related tweets that may be associated with the proximity and severity of floods. While Robinson & al. [125] presented a notification system for identifying earthquakes, but also, in first-hand reports published on Twitter from tweets of target areas in Australia and New Zealand.

On the other hand, with the reinforcement of the use of microblogging platforms such as Twitter during the sudden onset of a natural or anthropogenic disaster, thousands of disaster-related messages posted. These online messages contain important information that can also be helpful in making quick decisions to help the affected community if they are dealt with quickly and effectively [11, 39]. Many types of processing techniques ranging from machine learning to natural language processing through computational linguistics have been developed [58] for different purposes [64]. However, fully exploiting this data, despite the existence of some resources, such as annotated data and standardized lexical resources [121, 122]. Terpstra & al. [129] present a real-time analysis of Twitter data. But they use predefined geographic displays, the content of Twitter messages, and so on to analyze and filter disaster information.

Most event detection methods are based on keywords [59], such as Imran & al. [123], which analyze keywords used in tweets during catastrophic events to classify messages as real-time event reports, using a support vector machine (SVM). This related research includes the methodology of Vieweg & al. [124] and disaster ontology to identify also tweets that provide only situational awareness. Kongthon & al. [40] develop classifiers to analyze only the tweets of the floods in Thailand in 2011. Also, Starbird & al. [22] and Vieweg & al. [29] analyzed the use of the microblog and the life cycle of disasters [123].

Bella Robinson & al. [125] developed a notification system alerting earthquakes, but, from, only, first-hand reports published on Twitter. João Porto de Albuquerque & al. [52] examined Twitter platform microblogging messages (*tweets*) produced during the flood of the Elbe in June 2013 in Germany, but, using only, statistical analysis to identify general spatial patterns in the occurrence of flood-related tweets that may be associated with the proximity and severity of floods.

Our approach is most closely related to that of Koustav Rudra & al. [126], who studied Community Tweets, that are, abusive messages targeting specific religious / racial groups displayed during natural disasters. Except that they proposed methods

to counter them by using anti-communal tweets published by some users during catastrophic events.

TABLE 2.4 – Comparative table of all techniques and methods used in Alert Models

Approaches	Methods	Identification	Used OSN	Real-time analysis
[11]	/	Learning	Twitter	/
[52]	Statistical analysis	Identifying spatial patterns	Twitter	/
[59]	Based on keywords	/	Twitter	/
[121]	Annotated data	Hybrid crowdsourcing Machine learning	Twitter	Real-time
[122]	/	Learning	Twitter	/
[123]	Based on keywords	Classifying messages	Twitter	Real-time
[125]	Annotated data	A Notification system alerting earthquakes	Twitter	/
[126]	Annotated data	Identifying Community Tweets (abusive messages)	(Tweets) Twitter	/
[129]	Predefined geographic displays	/	Twitter	Real-time
[130]	Annotated data	/	Twitter	/

2.2.1.2 Functioning of neural learning

Figure 2.3 shows the architecture of the neural network that functions according to all the functionalities presented in Figure 2.2.

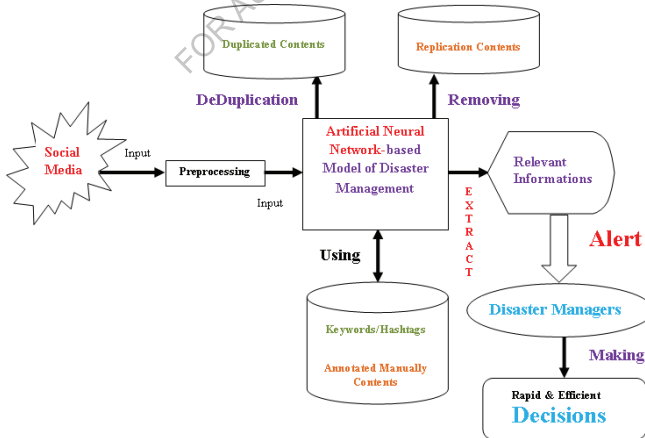


FIGURE 2.3 – Artificial Neural Network Architecture.

The perceptron is an algorithm for binary classifiers' supervised learning. A binary classifier is a function which can decide whether or not an input, represented by a

TABLE 2.5 – Classification of Neural Learning architectures.

Type	Architecture	Model - Training - Algorithm - Appli- cation	Ref
NL	Neural Network / Machine Learning	Discriminative- Supervised-Gradient Descent based Backpropagation- Classification	[14, 15]

vector of numbers, belongs to some specific class, making its predictions based on a linear predictor function, that combines a set of weights with the feature vector. Figure 2.4 shows the Automated Neural Environmen Platform.

Whether it's Twitter, Facebook or other social networks, these are platforms where people often express emotions. We detected emotions using four types of features : 1) interjections, 2) blasphemy, 3) emoticons, and 4) the general feeling of the message [128]. Interjections, blasphemies and emoticons are widely used by individuals to convey emotions such as anger, surprise, happiness, etc. To identify these three types of features, we used a combination of tags in the English tagger (*which contains tags for interjections, emoticons, etc.*), compiled lists of interjections and blasphemies on the Web for English, Spanish and French and regular expression patterns for emoticons [131].

Messages from Indonesia's target regions are checked for frequency bursts of disaster keywords and processed to identify evidence of a disaster. The advantage of our Disaster Alert model is that it relies on evidence from Twitter's and Facebook's first-hand reports. This multi-view template associates an input document vector with a multi-view space, based on neural network, whose contextual information is available.

2.2.2 Network Propagation Scheme

Having identified the relevant information, we have proposed different ways to alert public opinion but especially disaster managers so that they can quickly make effective decisions that can save lives. Figure 1.4 shows the flow chart of the network propagation scheme of the alert.

2.2.3 Performance Evaluation

We took into account the contents posted during catastrophic events summarized in Table 1.1. Note that the events (1), (2), (3), (4), (5), (6), (7) , (11), (12) and (13) are natural disasters and events (8), (9), (10) and (14) are anthropogenic disasters. In addition, we took into account events occurring in different geographic areas so

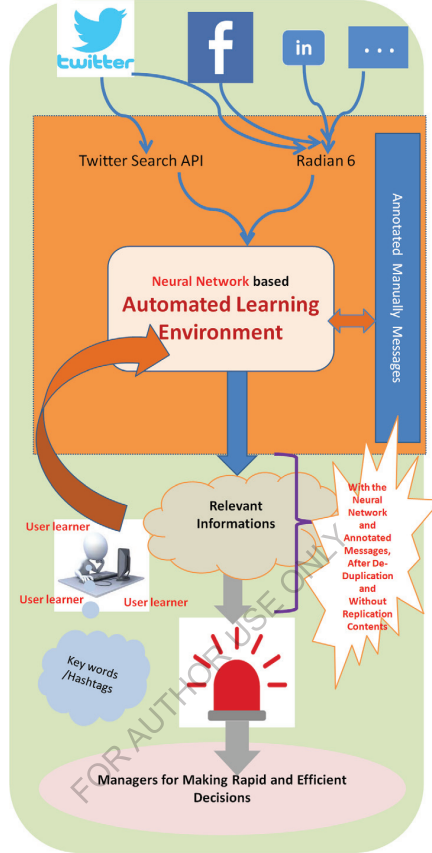


FIGURE 2.4 – Automated Neural Environmen Platform.

that this document is not influenced by any type of demographic data. The contents will be available to the community of potential researchers. We applied end-to-end key matching (API) search in Twitter and Facebook at each event according to [78]. For example, for the contents related to the event (2) of the earthquake of Haiti of January 12, 2010, we are looking for content with keywords such as *earthquake*, *Haiti*, Haiti earthquake for the messages of Facebook and #HaitiEarthquake, #Haiti and #Earthquake, for tweets from Twitter. For each word, we collected all the tweets returned by Twitter. We only consider contents in English for Facebook or Twitter. For each event, we report the number of content (*tweets and messages*) In order to show the effectiveness of our proposed model, we examined a specific event - the Indonesia

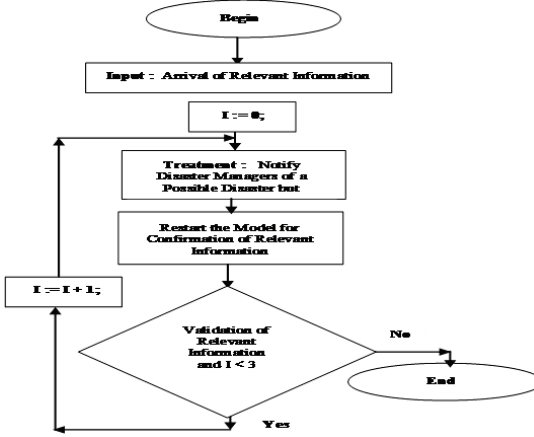


FIGURE 2.5 – Flow chart of the network propagation scheme of the alert.

earthquake in September 28th, 2018 and the Japan flood in July 2018 - and post-event messages on two social media : Twitter and Facebook.

The tweets were collected using the Twitter Search API using the #indonesia and #earthquake hashtags for the September 28, 2018 earthquake in Indonesia and the #Japan and #floods hashtags for floods. July 2018 in Japan. Facebook posts have been collected using the Radian6 tool in real time, as it allows. Facebook content was obtained using the search terms *Indonesia* and *earthquake* for the 2018 Indonesia earthquake and the *Japan* and *flood* keywords for the 2018 Japan floods.

With the neural network and after de-duplication, we have obtained the results illustrated on Table 2.6. All experiments reported here were performed on these two sets of data.

TABLE 2.6 – Examples of Contents Obtained, for a Set of Keywords or Hashtags, After De-Duplication.

Annotated	Indonesia	earthquake	Japan	floods
Contents	Twitter (Tweets)	Facebook (Contents)	Twitter (Tweets)	Facebook (Contents)
Contents	51,109	76,338	36,077	12,325

Standard relevance queries and judgments : we identified a set of disaster-specific information needs, as proposed in [52]. It's a set of keywords and hashtags for Facebook and Twitter respectively. Data collection and filtering are at the heart of disaster management using social media. The algorithms used to warn and alert depend on

the messages published on the social media and their quality. Thus, every effort must be made to maximize the number of messages relevant to the disaster and to remove non-informative messages as proposed in [132]. These contents are manually tagged (*annotated*) to remove those not related to the disaster (*non-informative messages*).

To understand the models and evaluate our proposed model for extracting relevant information through semantically-based learning, we need an annotation for a set of contents. We randomly sampled 1366 messages (*with 547 Facebook content and 819 tweets*) after deleting the duplicates, as reported in 2.6. These messages were observed independently by ourselves. The goal was to identify all relevant content relative to the predefined keywords. Table 2.7 presents the number of relevant documents found for each keyword or hashtag from the two datasets.

TABLE 2.7 – Examples of Relevant Content for a Set of Keywords or Hashtags.

Annotated Contents	Indonesia	earthquake	Japan	Floods
	Twitter (Tweets)	Facebook (Contents)	Twitter (Tweets)	Facebook (Contents)
Anthropogenic Disaster	55	59	18	32
Natural Disaster	289	192	255	268

This content, obtained in this table, manually labeled (*annotated*) and cleaned from non-informative messages, will be used, in this neural network based space, to analyze live new messages through the learning allowed by this neural network based space, enriched by relevant messages annotated manually.

As we want our alert model to be event - independent, it must be able to be used directly on content posted at later events. We therefore adopt the approach of using a set of features for the content extraction task. And since we have a large number of data sets (*14 sets of data*), a few will be dedicated to training and others to check the performance of our proposed model.

We compare the performance of the proposed set of features in two scenarios :

1. Classification in the field : where the classifier is trained and tested with content related to the same event using cross-validation and
2. Classification inter-domains : where the classifier is trained in the contents of an event and tested in another event. In this case, all the annotated contents of a particular event are used to form / develop the model, and then it is tested on all the contents of the rest of the events.

2.2.3.1 Model Selection

The performance of this system strongly depends on the selection of the appropriate model. We are now trying to select the most appropriate model for

our proposed feature set based on specific criteria. We consider four advanced classification models for the feature set above :

1. Random Forest (RF) ;
2. Logistic Regression (LR) ;
3. Naive Bayes (NB) ; and
4. Rule Based Classifier (RL)

We follow a simple approach - if any of the features mentioned above are present in a content, we mark it as relevant. Our neural network-based alert model is applied to all these pre-prepared data models. To judge the performance of these data models on the feature sets mentioned above, we have defined the following evaluation criteria. Each criterion is calculated and averaged over the different sets of training data.

1. Average accuracy in the domain : average accuracy of our alert model for the different events of the learning set, as in the scenario in the domain.
2. Medium inter-domain accuracy : average accuracy of our alert model in different inter-domain scenarios among the different events in the learning set.
3. Average accuracy for content : the detection of content with high accuracy is a necessary condition for our alert model. Therefore, we consider the average accuracy in training datasets.
4. Average F score for the contents : the F score of the system indicates the balance between coverage / recall and accuracy / precision.

Table 2.8 shows the Score Assessment Parameters for Ranking Models Using the Proposed Characteristics of the results shown in table 2.7.

TABLE 2.8 – Score Assessment Parameters for Ranking Models Using the Proposed Characteristics.

Classifier	Precision in the Domain	Inter-Domain Precision	Pre- Precision	F-score
Random Forest	0.9042	0.9304	0.9324	0.9258
Logic Regression	0.9135	0.8919	0.9113	0.8941
Random Forest	0.9254	0.9117	0.9509	0.9112
Random Forest	0.9308	0.9307	0.9518	0.9291

2.3 Conclusion

This chapter represents the first attempt to extract, in real time, information on catastrophic events from multiple sources on the Web (*Facebook and Twitter*) and to

immediately alert disaster managers so that they can make quick and effective decisions that could, perhaps, save lives. Indeed, we propose a new ad hoc real-time alert model for the management of disaster, whether natural or anthropogenic, based on a new multi-view recovery model from multiple sources. Such an approach is really useful for local disaster monitoring, but also for helping to make appropriate decisions.

1. We have developed a neuron-based model that uses low-level content learning capabilities to automatically separate relevant information from redundant (*eg. retweets for Twitter*) or abusive information (*at the level of the community, for example*).
2. This model, based on a neural network, uses the learning capabilities of multiple source content (*here Twitter and Facebook*) to automatically retrieve relevant information using a set of keywords and hashtags respectively, related to various catastrophic events, whether natural or anthropogenic.
3. Keeping in mind the limitations of the previous work [14], we have developed an event-independent alert model that can be used directly to filter content from multiple sources at future events. Experiments on multi-disaster-related content flows with diverse characteristics show that our proposed model outperforms vocabulary-based approaches. While our approach filters content (*Twitter tweets and Facebook posts at once*).
4. Once we developed this real-time alert model, we immediately tested it on the two catastrophic events of late summer, namely the Japan Floods from June 28 to July 9, 2018 with 2,000 dead, 2,500 wounded and 330,000 homeless and the earthquake of Haiti of October 6, 2018 with 18 dead and 548 wounded.
5. We have also proposed a warning network propagation scheme that will enable disaster managers to make effective decisions quickly and effectively to save lives.

"Every great advance in science has issued from a new audacity of imagination."

John Dewey

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Chapitre 3

Deep Learning-based Automated Learning Environment through Social Media for Disaster Management

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3.1 Introduction

IN recent years, the frequency of large-scale disasters has increased worldwide. The Boumerdes earthquake [1] of May 21, 2003 and the Chlef earthquake (Algeria), for example, which occurred in 1980 (7.3 degrees on the Richter scale) caused 2,633 deaths and caused the collapse of 20,000 homes. Every time a disaster occurs, many recognize the importance of disaster management. The majority of disasters have huge and lasting impacts on human lives and societies. In order to minimize the consequences of such events, it is essential to prepare and develop resilient communities by educating them with the appropriate and up-to-date information needed to cope with disasters. We will examine here how individuals, organizations and communities develop their skills in response to unforeseen events, whether in preparation, in disaster management or in the institutional aspect.

Our analytical framework is guided by the important social phenomenon that occurs in almost every disaster and the vastness of activity that brings this observation of evidence to wonder. Disasters such as earthquakes, floods, explosions, terrorist acts and tsunamis result in catastrophic human suffering, loss of property and other negative consequences [2].

People use social networking to post situational updates in various forms such as text messages, images and videos. Numerous studies [3] have shown that this online information is useful for a quick response to disaster management. These analyzes of the use of social networking in disasters have identified a distinct role for local users of the event (or *in the field*), who are more likely to generate useful information to improve situational awareness [4].

Thanks to the improvement of storage technics (big data) and the ease of use, the participatory culture and the interoperability of end-users (Web 2.0), we are being confronted with information overload stored, notably the Web which is a popular and interactive way of disseminating information today. It is notably immense, diverse and dynamic. This new era is defined by the rapid development of knowledge and that traditional learning initiatives have not been able to cope with the growing complexity and rapid changes of the new knowledge society. Therefore, a new vision of learning is needed, namely Automated Learning. Its aim is to discover, in large amounts of data, valuable information that can help understand these data or predict the behavior of future data. This process involves examining large amounts of data (stored in warehouses or in streaming) using statistical, mathematical or eventually AI techniques. This knowledge, unknown at first, may be correlations, patterns or general

trends in that data. Automated Learning is based on the idea of analyzing problems to understand their principles and develop appropriate mathematical models. For that, experimental data are required to verify the correction of the system or the estimation of some difficult parameters to mathematical modeling [5]. A fundamental shift towards a more open and learner-centered learning model is needed where Automated Learning uses various manual and automatic tools : we start by describing the data, summarize their statistical attributes (means, variances, covariance), visualize them using the curves, the graphs, the diagrams, and finally look for the significant links potential between variables (such as values that are repeated). But the description of the data alone does not provide a plan of action. We must build a prediction model based on the information discovered, and we validate this model on other data. Over the past decade, it has been argued that Automated Learning could meet the needs of the new knowledge society and transform learning. However, in the majority of cases, systems do not have principles understood or are too complex for mathematical modeling. Automated Learning, which suggests complex and unimaginable specific impacts on the society and more broadly the economy, remains primarily in the field of computer science, but is closely linked to cognitive science, neuroscience, biology and psychology and could lead to the crossroads of systems, nanotechnology, biotechnology and cognitive science, leading to artificial intelligence with a wider base.

Smart Disaster education plays a key role in encouraging community members to develop disaster management skills. In addition, these members also share their knowledge and help one another in disaster preparedness, like with other disaster education techniques, namely, Evacuation drills, Games Based-Evacuation Drill [6], Paradigmatic Tourism [133], Dark Tourism, Penumbral Tourism, Mixed Realistic Games or Alternative Realistic Games (ARG) [8] and tower defense game [9], or other evacuation techniques as Simulations.

In this research work [16], we present the theoretical, design, implementation and evaluation details of an environment, a learning framework whose main objective is to help users become familiar with all of these different learning models using a wide variety of media and digital data available in this environment. This automated learning environment is fundamentally personal, social, distributed, ubiquitous, flexible and dynamic.

3.2 Smart Interface-based Automated Learning Environment Through Social Media for Disaster Management and Smart Disaster Education

This section consists of studying Automated Learning, Disaster and Disaster Management, Social Networking and Disaster Education.

3.2.1 Improving Automated Learning

Artificial intelligence (AI) is a combination of reinforcement learning (RL) and deep learning (DL) [104], represented mathematically, as(see Figure 2) :

$$AI = RL + DL \quad (3.1)$$

where : AI represents Artificial Intelligence, RL represents Reinforcement Learning, and DL represents Deep Learning.

3.2.2 Machine learning (NN)

Machine Learning is an artificial intelligence technology that enables computers to learn without having been explicitly programmed to do so. To learn and increase, however, computers need data to analyze and train on.

While machine learning is not new, its precise definition is still confused for many people. Concretely, it is a modern science for discovering patterns and making predictions from data based on statistics, data mining, pattern recognition and predictive analyzes. The first algorithms were created at the end of the 1950s. The best known of these is none other than the *Perceptron*.

The perceptron is an algorithm for supervised learning of binary classifiers. A binary classifier is a function which can decide whether or not an input, represented by a vector of numbers, belongs to some specific class. It is a type of linear classifier, i.e. a classification algorithm that makes its predictions based on a linear predictor function combining a set of weights with the feature vector. The perceptron algorithm was invented in 1958 at the Cornell Aeronautical Laboratory by Frank Rosenblatt [134], funded by the United States Office of Naval Research.[14, 15]

3.2.3 Deep learning

In recent years, Deep learning methodologies have achieved impressive results in computer vision [135], speech recognition, image processing, and handwritten recog-

nition of characters, while they is currently in its infancy in fault diagnosis [136]. It can also overcome limitations of shallow networks that prevent efficient training and abstractions of hierarchical representations of multi-dimensional training data [136] in many instances. Deep learning offers a set of units such as [137] convolution unit, [138] recurrent unit, and long-short term memory unit for feature extraction on samples with distinct features, delving into the math behind training algorithms used in recent deep networks, explaining existing shortcomings, [136] improvements and implementations. The paper also covers different types of deep architectures corresponding to the two types of learning such as Feedforward and Feedbackward neural networks.

3.2.3.1 Discussions about Deep Learning Models

TABLE 3.1 – Classification of Feedforward Neural Network (FFNN) architectures with Model, Training, Algorithm, Objective and Limitations.

Architecture	Model - Training - Algorithm - Application	Limitations	Ref
FFNN	Supervised-Gradient Descent based Backpropagation- <i>Binary classification</i>	No Extrapolation [139]	[16, 17, 140]
ConvNets (CNN)	Discriminative-Supervised-Gradient Descent based Backpropagation- <i>reducing frequency variations; Image recognition / classification</i>	Temporal modeling-No increasing validation precision with layers stacking-not encoding objects position and orientation [143, 144]	[18, 47, 141, 142]
ResNets	Discriminative-Supervised-Gradient Descent based Backpropagation- <i>Image recognition</i>	Increased complexity of architecture; Implementing Batch; Adding skip level connections	[16, 17, 140]
Auto-encoder	Generative-Unsupervised-Backpropagation- <i>Image recognition</i>	not discover slow modes [148]	[138, 145, 146]
Generative Adversarial Networks	Generative-Discriminative-Unsupervised-Backpropagation- <i>Generating realistic fake data-building 3D models or image improvement</i>	distribution learning poorly made ¹	[28]
Restricted Boltzmann Machine (RBM)	supervised or unsupervised-Generative with Discriminative finetuning-Unsupervised-Gradient Descent based Contrastive divergence- <i>Dimensionality reduction, feature learning, topic modeling, classification, collaborative filtering, body quantum mechanics</i>	difficult training; tricky partition function making computing log likelihood infeasible	[138]
Recurrent NN (RNN)	Discriminative-Supervised-Gradient Descent & Backpropagation through Time- <i>Natural Language Processing; Language Translation; Sequences recognition</i>	Difficult inference, of time series unsupervised problem, in negative time [147]	[138, 145, 146]
Bi-directional RNN	Discriminative-Supervised-Gradient Descent & Backpropagation through Time- <i>Natural language processing (NLP)</i>	Trained with input information limitation up to preset future frame [147]	[28]

Feedforward neural networks are not capable of extrapolation [139]. Disadvantages of ResNets are Increased complexity of architecture such as Implementation of Batch normalization layers since ResNet heavily depends on it and Adding skip level connections for which the dimensionality is essential between the different layers.

TABLE 3.2 – Classification of Deep Learning architectures with Model, Training, Algorithm, Objective and Limitations.

Type	Architecture	Model - Training - Algorithm - Application	Limitations	Ref
FBNN	LSTM	Discriminative-Supervised-Gradient Descent & Backpropagation through Time- <i>Natural Language Processing; Language Translation; Temporal data as stock market</i>	No obtaining well-defined temporal information [144, 149]	[138]
	Fully Connected-LSTM	Discriminative-Supervised-Gradient Descent & Backpropagation through Time- <i>Natural Language Processing; Language Translation; temporal modeling; Learning non-linear and complex processes</i>	No obtaining well-defined temporal information [144, 149]	[150]
	Bi-Directional LSTM	Discriminative-Supervised-Gradient Descent & Backpropagation- <i>Time-natural language processing and language translation</i>	Bad Presentation with multi-level features [143]	[141]
Radial Basis Fct NN	RBF Network	Discriminative-Supervised and Unsupervised- <i>Means clustering, Least square function, function approximation-time series prediction</i>	slow classification due to computing RBF function	[151]
Kohonen SO NN	Nodes arranged in grid	Generative-Unsupervised- <i>Dimensionality reduction, optimization problems or clustering analysis; Competitive Learning</i>	SOM algorithm Problems ²	[151]

Resnets, a CNN variant designed for Computer Vision image classification tasks, was proposed to overcome the problems of VGG styled CNNs, as just stacking convolutional (CNNs) layers to make the model deeper does not guarantee an increase in validation accuracy. ResNet architecture was introduced for skipping connections. While, after increasing the number of such ConvNets layers for a certain limit, the generalization ability and validation accuracy decrease. CNN do not encode the position and orientation of object and Lack of ability to be spatially invariant to the input data. Autoencoders have been proposed as a deep learning regressionbased approach to the discovery of slow modes in dynamical systems. However, a rigorous analysis of nonlinear autoencoders remains lacking. Autoencoders can not correctly discover slow modes and fail [148].

The generator produces a distribution Psynth supported on a small number of images, quite far from the distribution on real images, and yet the discriminator cannot meaningfully distinguish between the two distributions³.

GANs, with finite size discriminators and generators, may not do distribution learning very well, even though training may appear to have succeeded.

Restricted Boltzmann machine is applied algorithm used for classification, regression, topic modeling, collaborative filtering, and feature learning. RBM deals with the basic unit of composition which progressively grown into many popular architectures and used widely in many large scale industries. Restricted Boltzmann machine is used for neuroimaging, Sparse image reconstruction in mine planning and also in Radar

3. <https://simons.berkeley.edu/news/research-vignette-promise-and-limitations-generative-adversarial-nets-gans>

target recognition

RBM able to solve imbalanced data problem by SMOTE procedure RBM finding missing values by Gibbs sampling which is applied to cover the unknown values. RBM overcomes the problem of noisy labels by uncorrected label data and its reconstruction errors. The problem unstructured data is rectified by feature extractor which transforms the raw data into hidden units.

The primary disadvantage is that RBMs are tricky to train well since the common algorithm used, Contrastive Divergence, requires sampling from a Monte Carlo Markov Chain, requiring a bit of care to get things just right. Furthermore, that tricky partition function in the model's energy makes computing the log likelihood under the model infeasible (or rather intractable). So, it cannot even track the loss caring about (let alone take derivatives with respect to). In an autoencoder, in contrast, one can at least track cross entropy, being minimized by the model of learning algorithm-back (propagation of errors). The generator produces a distribution P_{synth} supported on a small number of images, quite far from the distribution on real images, and yet the discriminator cannot meaningfully distinguish between the two distributions.

When a deeper feedforward neural network is created, it can be giving the model the ability to capture more complex representations, as for Image recognition tasks (convolutional neural networks), Natural language processing (like how Google Translate works), Bio-informatics tasks such as figuring out gene relationships or Voice recognition tasks.

If it is tried to create a deep neural network to model more simpler phenomena, it is running the risk of over-fitting the data, losing the ability able to generalize to new examples and factoring the amount of resources it takes to train a deep neural net. When training deep neural networks it often have to use specialized hardware (i.e. a dedicated GPU) or cloud computing services (i.e. AWS, Google Cloud) to train the network effectively.

Two of the important variations for the LSTM model are deep LSTM (DLSTM) and convolutional LSTM (CLSTM). DLSTM differs from the general LSTM in the number of layers the model contains. A single-layer LSTM will not be able to obtain well-defined temporal information. However, when more layers are stacked in the LSTM model, it will be able to acquire better temporal features, and hence will be more suitable in capturing motion in the time dimension. In CLSTM, the data are first passed through convolutional layers, which ensure in capturing the spatial features. The output from the CNN is provided to the LSTM, which will get the temporal features, and hence, the model will capture a motion with respect to both space and time. These two variants can also be combined to give a convolutional deep LSTM, where the outputs from a CNN are given to a multilayer stacked LSTM, which is guaranteed to provide a better

result at the cost of increased computational complexity [149].

The BRNN can be trained without the limitation of using input information just up to a preset future frame, comparatively to RNN. As an example of an unsupervised task, gaps filling in high-dimensional time series with complex dynamics, where unidirectional RNNs have recently been trained successfully to model such time series, but inference in the negative time direction is non-trivial [147]. Both Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) have shown improvements over Deep Neural Networks (DNNs) across a wide variety of speech recognition tasks. CNNs, LSTMs and DNNs are complementary in their modeling capabilities, as CNNs are good at reducing frequency variations, LSTMs are good at temporal modeling, and DNNs are appropriate for mapping features to a more separable space. It is interesting to take advantage of the complementarity of CNNs, LSTMs and DNNs by combining them into one unified architecture [144].

ConvNets present some limitations of temporal modeling [143]. The Bi-directional Long Short-Term Memory (BLTSM) conventional models have encountered some limitations in presentation with multi-level features but can keep track of the temporal information while enabling deep representations in the data [143].

Deep Bi-directional Long Short-Term Memory (DBLSTM) architecture with multi-levels feature presentation for sentiment polarity classification (SPC) on social data. By using DBLSTM, It can exploit more level features than BLTSM and inherit temporal modeling in BLTSM [143].

Radial basis function (RBF) networks have advantages of easy design, good generalization, strong tolerance to input noise, and online learning ability. Although the training is faster in RBF network but classification is slow in comparison to Multi layer Perceptron due to fact that every node in hidden layer have to compute the RBF function for the input sample vector during classification.

Kohonen Self Organizing NN is very popular might be because it has an easy to understand algorithm, it is simple to use, and it produces good and intuitive results. They are inherent to the model of mapping data points from a high dimensional data-space onto a 2-dimensional map while preserving local distances. It can be overcome by using Somor Gtm combined with other data-mining methods such as pure clustering or multi-dimensional scaling techniques.

Abiodun et al. (2018) [104] recommend that future research can focus on combining various ANN models into a single network-wide application, as needed and depending on the characteristics of the various ANN models.

TABLE 3.3 – Classification of Social Media-based Recent Neural Learning Approches of Disaster Management.

Social Media-based Approches	References
Twitter	[28, 141, 47]
Twitter & Facebook	[14, 15]
All the Web	[16, 17]

3.2.3.2 Classification of Deep Learning Architectures

Neural learning is carried out by Feedforward or Feedback neural network. In Feedforward, we have supervised learning such as Feedforward neural network itself for classification [16], convolutional neural network [18, 47, 141] for image recognition/classification or Residual neural network (ResNets) [152] for image recognition, and unsupervised learning such as Autoencoder [138] for Dimensionality reduction and encoding, Generative Adversarial Network [28] network for generating realistic fake data, reconstruction of 3D models or image improvement and with supervised or unsupervised learning such as Restricted Boltzmann machine [138] for dimensionality reduction, feature learning, topic modeling, classification, collaborative filtering or many body quantum mechanics. In Feedback, we have only supervised leaning such as Recurrent neural network [138] for sequences recognition as precise timing, Bidirectional Recurrent Neural Network [28] for natural language processing (NLP), Long Short-Term Memory [138] for temporal data such as stock market values over a period of time, video frames, Fully Connected-LSTM [150] for learning non-linear and complex processes in hydrological or meteorological modeling and Bi-Directional-LSTM [141] through time-natural language processing and language translation. Neural learning can be trained in either supervised/unsupervised ways by Radial Basic Function Network [151] for M-means clustering, Least square function, function approximation and time series prediction or unsupervised ways by Kohonen Self Organizing Network [151] for dimensionality reduction, optimization problems or clustering analysis.

3.2.4 Improving Social Networking

Social Networking plays a very important role in disaster management. Collecting and classifying the content (information and images) posted on social networking platforms, to help humanitarian organizations to become aware of the situation and to launch relief operations, understanding these voluminous and high velocity data is a difficult task which requires the following Crowdsourcing, Crowdtasking and Collaborative Disaster Management.

— Collaborative Disaster Management

The catastrophe is a situation that requires quick and effective collaborative decisions, with a high degree of uncertainty and complexity [153].

— **Crowd-sourcing and Crowd-tasking in Disaster Management**

Crowd-tasking describes the management of volunteers applying new media and the Crowd-Tasking System (CTS) is defined as combination of technical solutions (crowdtasking applications, CTA), task managers, task management software in the situation room and volunteers themselves [19].

The rise of social networks has enabled ad hoc citizen groups to organize large-scale activities in a flexible way. In post-disaster relief operations, it is assumed that all fixed cellular infrastructures do not be functional. The adaptability of Peer-to-Peer (P2P) networks must be exploited to respond to the characteristics of disaster situations. P2P networks have been used to interconnect field workers by just one active connection between a peer and the control room to perpetuate the disaster management system. Thus, the benefits of applying the P2P paradigm have applied to the Geo-Collaboration for Disaster Management as an alternative to the most common Client/Server approaches of GIS and propose to interconnect mobile operators through P2P networks. Mobile P2P systems are a relatively new approach to combine P2P concepts with mobile computing technologies such as WirelessLAN, Bluetooth and telecommunication networks such as GSM, GPRS or UMTS. Mobility support is becoming an increasingly important research topic in the design of P2P systems, especially in Disaster Management. As a result, users' flexibility and independence can also be increased in a physical way, so that they can interact with P2P-based mobile anytime, anywhere mostly in Disaster Management. The study found that New Trends in Scientific Research in Disaster Management focus on two broad trends, namely [154] prediction and monitoring for early warning, and Relevant information retrieval and situational awareness classification.

Models for retrieving information from multiple sources

In a bilingual recovery, comparable document-aligned data [98] is used to easily learn patterns of common topics or word concepts. However, the preparation of comparable documented data requires a lot of human involvement and time. Therefore, such approaches are not appropriate when information about an ongoing event should be retrieved quickly. The work of Joseph A. Shaw and Edward A. Fox [99] focuses on methods for recovering separately from different sources and then combining them using data fusion techniques. Multi-view recovery based on deep learning that attempts to learn document embedding in a common space, where differences between different data sources would not exist. Unfortunately, this approach has low performance scores for all the methods used : indicating that the problem is difficult and requires bet-

ter methods. According to Cynthia D. Balana [100], Google Crisis Response's (GCR) National Risk Management and Disaster Response Board (NDRRMC) had integrated Facebook and Twitter into its surveillance system and website for a quick response.

TABLE 3.4 – Comparative table of all techniques and methods used in Models for retrieving information from multiple sources.

Approach	Exigences	Real-Time	Multiple Sources	Method
[78]	Low performance scores	/	/	Multi-view recovery based on deep learning
[98]	Requires a lot of human involvement and time	Not appropriate for real-time approach	/	Learning patterns with comparable documents
[99]	/	/	Recovering separately from different sources	Combining using data fusion techniques
[100]	/	/	Surveillance System Facebook and Twitter	/
Proposed Approach	Annotating manually information	Appropriate for real-time	Multiple sources (Facebook Twitter)	Learning with Neural Network

Whether it's Twitter, Facebook, Viber, Messenger, any forum or any thing in the Web, these are platforms where people often express emotions. Emotions are detected using four types of features, namely : interjections, blasphemy, emoticons, and the general feeling of the message.

Interjections, blasphemies and emoticons are widely used by individuals to convey emotions such as anger, surprise, happiness, etc. To identify these three types of features, a combination of POS tags is used in the English tagger (which contains tags for interjections, emoticons, etc.), compiled lists of interjections and blasphemies on the Web for English and regular expression patterns for emoticons.

Table 3.5 shows a comparison of all techniques and methods used in Online Listening and/or Monitoring.

Our proposed model is based on a semi-supervised inductive technique to use unlabeled, multi-source data, which is often abundant during a crisis event, with less data previously labeled than previous events (*Table 2 and Table 4*).

Online Listening and Monitoring

Contents were collected from all online channels tracked automatically by the Online Listening Tool, namely Radian6 [92], from websites to social media : Twitter, Facebook, LinkedIn, Instagram, Google+, Youtube and so on. Radian6 tool or any its competitor³, such as Awario, Brand24.com, Brandwatch, Mention, Keyhole, Socialert.net, SocialPilot.co, Simplify360, etc. of Online Listening or Monitoring tools, is

3. <https://www.quora.com/What-are-the-top-competitors-to-radian6-I-am-looking-for-alternate-measuring-tool-to-use-with-my-clients>

TABLE 3.5 – Comparative table of all techniques and methods used in Online Listening and/or Monitoring.

Approach	Tool	Methodology	Used OSN
[78]	Twitter APIs for tweets and Radian6 for Facebook messages	Based on keywords-Annotated manually messages, harvesting contents, cleaning them and extracting relevant information	Twitter & Facebook
[14] : our previous work	Twitter APIs for tweets and Radian6 for Facebook messages	Based on keywords-Annotated manually messages, , harvesting contents, cleaning them and Learning Relevant information with neural network, in <u>Real-time</u>	Twitter & Facebook
Proposed Approach	Radian6 for all contents	Based on keywords-Annotated manually messages, , harvesting contents, cleaning them and Learning Relevant information with neural network, in <u>Real-time</u>	

TABLE 3.6 – Comparative table of all techniques and methods used in Models for retrieving information from multiple sources.

Approach	Exigences	Real-Time	Multiple Sources	Method
[78]	Low performance scores	/	/	Multi-view recovery based on deep learning
[98]	Requires a lot of human involvement and time	Not appropriate for real-time approach	/	Learning patterns with comparable documents
[99]	/	/	Recovering separately from different sources	Combining using data fusion techniques
[100]	/	/	Surveillance System Facebook and Twitter	/
Proposed Approach	Annotating manually information	Appropriate for real-time	Multiple sources (Facebook Twitter)	Learning with Neural Network

used to collect information, using the search keywords 'Boumerdes' and 'earthquake' for the Boumerdes earthquake and the 'Algiers' and 'flood' keywords for the Algiers floods. Online listening tools provide the model, which reasonably represents the essentials, namely :

1. Harvesting contents : (such as conversations at the social media, news or any information in the Web) ;
2. Cleaning the data of duplication and replication content : eliminating, from the content, any dubbed information like re-tweet, and any harmful or redundant information ;
3. Enabling relevance : thanks to the neural learning, obtaining of relevant information using the corpus. This corpus is the annotated manually content which

is realized by volunteers ;

4. Analyzing the results : During this stage, the verification and analysis of the results is carried out in order to ensure adequacy so that it is obtained to alert the disaster managers.

It is more the limitation of freedoms by monitoring the attitudes of entire populations that I consider the main disadvantage of the online tools of listening and monitoring. As for the benefits, they are huge. For example, finding some details of an ultimate discussion is one of them. Being able to predict and prevent the outbreak of a disaster and make certain decisions that could save lives, has no commercial value. It is above all a value of great morality.

3.2.5 Related works

In 2009, Tobita et al (2009) [32] have initied by introducing an Integrated disaster simulator using Web GIS and its application to community disaster mitigation activities. In 2012, Fischer and al. [8] introduced Dark Tourism, Penumbral Tourism, Mixed Realistic Games or Alternative Realistic Games (ARG). In 2015, Tsai et al (2015) [9] introduced a flood protection computer game for disaster education. Followed by Mitsuahara et al (2015) [133] developed a Game-based evacuation drill using real world edutainment. Finally, in 2016, Kawai et al (2016) [6] developed a Tsunami Evacuation Drill System using Motion Hazard Map and Smart Devices. While Power and Robinson (2016) proposed Comparing Felt Reports and Tweets About Earthquakes.

3.3 Automated Learning Environment in Disaster Management

Artificial Neural Networks (ANN), inspired by biological neural networks, are like the brains of animals. Such systems *learn* to perform tasks by considering examples, usually without being programmed, with task-specific rules. A neural network is based on a collection of connected units or nodes called neurons, which model the neurons of a biological brain. Each connection, like the synapses of a biological brain, can transmit a signal from one artificial neuron to another.

3.3.1 Automated Learning Environment : modeling overview

Learning by retro-propagation is a method of gradient descent : in a given iteration, the direction of the search is given by the negative gradient of energy, while the pitch in this direction is given by a constant (velocity learning) chosen by the user. However, it

is well known in the optimization literature that pure methods of gradient descent tend to be very inefficient. While keeping its great advantage, namely obtaining a speed of convergence faster around the global optimum and, at the same time, a convergence accuracy that can be higher.

The feedforward artificial neural networks (ANN) are currently used in various applications with great success. Their first advantage is that they do not require a userspecific problem solving algorithm (as in the case of classical programming), but they learn from examples, much like human beings . Their second advantage is that they have an inherent capacity for generalization. This means that they can identify and respond to patterns similar to but not identical to those with which they were formed. On the other hand, the development of a feedforward ANN model also poses some problems, the most important being that there is no guarantee that the model will work well for the problem to be solved [155].

The entire ANN modeling procedure has been studied with the aim of introducing systematic methods leading to always efficient ANN models, namely the collection of learning data, preprocessing and post-processing of data, different types of activation functions, initialization of weights, learning algorithms and error functions. Although all of these factors affect ANN's performance, increased attention has been focused on finding the best architecture. This is justified not only by the fact that it is directly related to the performance of the model, but also by the fact that there is no theoretical background on how this architecture will be found or on its appearance [155]. The most typical method followed is a repetitive trial and error process, in which a large number of different architectures are examined and compared to each other. Therefore, this process is time consuming and relies mainly on the experience and intuition of the human expert, which implies a high degree of uncertainty. Nevertheless, we cite four different approaches, namely :

- The empirical or statistical methods used to study the effect of internal parameters of an ANN and choose appropriate values depending on the performance of the model [156],
- The hybrid methods such as fuzzy inference [157],
- The constructive and / or pruning algorithms [158] and finally
- Tthe evolutionary strategies.

The most systematic and general of these methods uses the principles of Taguchi's conception of experiments [159]. The best combination of the number of hidden layers, the number of hidden neurons, the choice of input factors, the parameters of the training algorithm, etc. can be identified with these methods even though they are usually case-oriented.

In the hybrid methods [157], the ANN can be interpreted as an adaptive fuzzy

system or can operate on fuzzy numbers instead of real numbers. In the constructive and / or pruning algorithms that, respectively, add and / or delete neurons of an initial architecture using a previously specified criterion to indicate how the ANN performance is affected by the changes [158]. The basic rules are that neurons are added when training is slow, and that neurons are suppressed when a change in the value of a neuron does not match at a change in the response of the network or when the weight values associated with this neuron remain constant during a large number of training periods.

In the evolutionary strategies, the topological space is explored by varying the number of hidden layers and hidden neurons by applying genetic operators and evaluating different architectures according to an objective function. Since the constructive and pruning algorithms are essentially gradient descent methods, their convergence towards the global minimum is not guaranteed and can therefore be trapped at a local minimum close to the point of the search space from which algorithms has been launched.

An artificial neuron that receives a signal can process it and then report other artificial neurons connected to it. In current implementations of neural networks, the signal at a connection between artificial neurons is a real number and the output of each artificial neuron is computed by a nonlinear function of the sum of its inputs. The connections between artificial neurons are called *edges*. Neurons and artificial edges usually have a weight that adjusts as learning progresses. The weight increases or decreases the signal strength during a connection.

Deep learning allows computer models composed of multiple layers of processing to learn data representations with multiple levels of abstraction : this has greatly improved the state of the art in speech recognition, visual recognition and objects, object detection and many other areas such as bioinformatics, Sequential Pattern Mining or genomics [37].

3.3.2 Configurations of the Neural Network Parameters

Problem : how to determine the coefficients w_0 , w_1 and w_2 ? To the Error function : If $y = 1$, we want to have $p(x)$ as large as possible. We define the error in this case by $-\ln p(x)$.

Symmetrically, if $y = 0$, we want to have $p(x)$ as small as possible. The error is then $-\ln(1 - p(x))$. So, the general formula (3.2) is :

$$error = -y * \ln(p(x)) - (1 - y) * \ln(1 - p(x)) \quad (3.2)$$

Figure 3.1 shows that a given neuron in a layer k consists of elements such as input

flux of all precedent neurons and output flows towards all the following neurons with their respective weights.

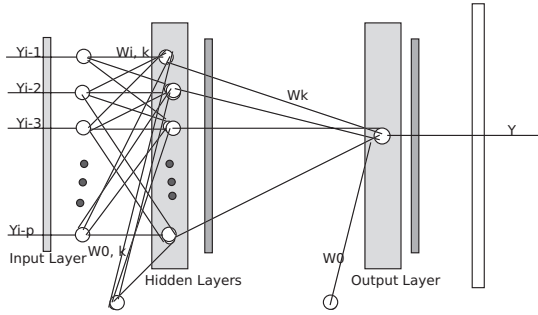


FIGURE 3.1 – Neural Network Structure.

Figure 3.2 shows Feedforward Neural Network Architecture.

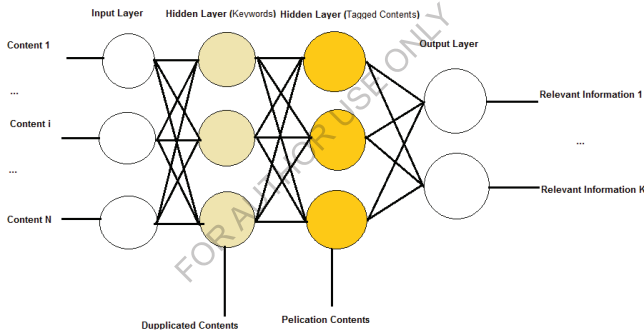


FIGURE 3.2 – Feedforward Neural Network Architecture.

Minimization of the error : Once an error function has been defined, the problem of learning becomes an optimization problem : find the coefficient vector w^* which minimizes the error. In the case of logistic regression, this vector is unique because the error function is convex. The solution vector w^* can be obtained by an iterative algorithm.

Probability of classifier error : Once the optimum w^* coefficient vector is determined, a program (classifier) is available to classify a new individual. To estimate the error probability of the classifier, it is necessary to have an independent test set.

The basic version of frequent pattern extraction is used to search a table in a Content Set whose values are Booleans indicating the presence or absence of a property. O is a finite set of objects. P is a finite set of properties. $\mathcal{R}$ is a relation on $O \times P$

which makes it possible to indicate if an object x has a property p (noted $x\mathfrak{R}p$) or not. For example, in the case of basket analysis in a supermarket, O is the set of purchase transactions, P is the set of items and $\mathfrak{R}$ is the relationship indicating whether an item is purchased in transaction t .

3.3.3 Manipulating of the Automated Learning Environment

We used the Model-View-Controller (MVC) model for event management design. We have also extended it to data mining specifically to the Sequential Model Exploration that we can apply to Marketing for example.

Figure 3.3 shows the launching of Feedforward Neural Network (Automated Learning Environment).

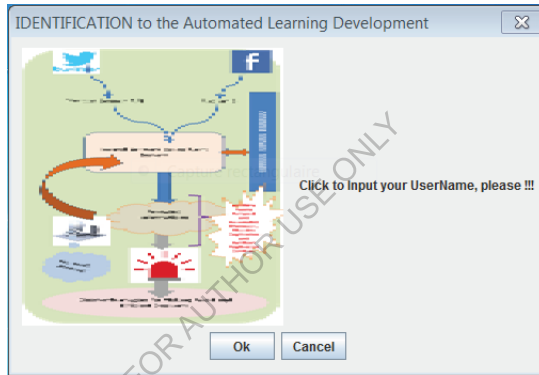


FIGURE 3.3 – Identification in the Automated Learning Platform

We used the Apriori algorithm used in Knowledge Discovery in Databases (KDD). It has been improved by Savasere [160]. We have adapted it to streaming in order to use the data of social networks. We have also used and adapted the DIC algorithm of [161] to streaming. Figure 3.4 shows the Architecture of Disaster Management Platform.

3.3.4 Smart Disaster Education

Disaster education plays a key role, first, in helping citizens to become aware of the disaster dangers, and then, in encouraging them to develop skills in disaster management. This environment is designed to support an introductory traineeship in disaster management for trainees and future disaster managers. Community members usually share their knowledge and help one another in disaster preparedness. Evacuation drills often take place in schools, businesses and communities [6], [133, 9, 32, 37, 48, 133].

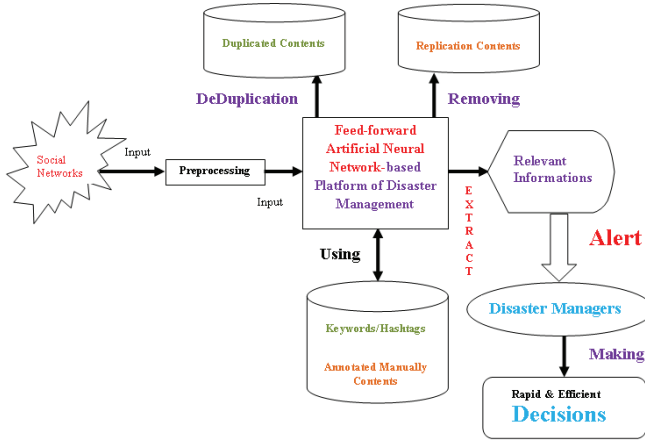


FIGURE 3.4 – The Automated Learning Environment

This environment is designed to support an introductory traineeship in disaster management for citizens, trainees and future disaster managers. The trainee can use this tool in the following modes :

- 1 Novice learning environment. To work with this tool, the trainee can use a complete set of automated design and learning tools. One of the functions of the environment is to display visualizations of the trainee's programs. With the help of the environment, the trainee can observe various programs "at work", experimenting them and gradually learning from his experience, observations and mistakes.
- 2 Learning environment for beginners. At any point in the job, the trainee can ask this environment to generate (move on) the next step : a new concept or a new build, a build example, a test, an example of a problem solver or a problem to be solved. This tool analyzes domain knowledge and the trainee's model and provides both the optimal stage and a list of all relevant operations (ie, ready to be applied). The trainee who is not satisfied with the optimal operation suggested by the system can choose any appropriate operation using adaptive hierarchical menus.
- 3 Online manual rehearsal. At any time during the work, the trainee has a menu to access all previous courses : presentation of any previously learned concept, demonstration of all the examples learned and analysis of any problem explained or resolved. This mode provides access to the material learned from the course as a reference, thereby supporting example-based online help.

3.3.5 Manipulating of the Automated Learning Environment

The trainee, once registered, with his own username and password, and thus defines his own learning environment, can thus launch the learning environment, already configured, to extract content from social media, analyze it. The keywords and manually annotated contents are already operational : they can view with the *Viewing* menu. Listening and Monitoring is programmed to *All the Web*. Processing is set to *Streaming*, the alert to *Automatic* and Learning to *Neural Learning*.

The novice can simply follow the progress of this treatment by viewing the relevant content, the applied content and the replication content using the *Viewing* menu.

In the event that he wants to participate (intervene) au processus, he can, for example, change the type of Learning with the *Learning* menu, change Listening and Monitoring, the treatment or the alert.

It can also intervene by adding other keywords in the *Updating Files / Updating Keywords* menu. He can also modify the alert to make it manual : thus, it will be up to him to alert.

If he is interested in understanding how this tool works, he selects *Content Treatment/Treatment* menu. Thus, it will be up to him to insert a text in the *Updating Files / Inserting Content* menu.

Thus, he can follow his treatment with the *Content management* menu for the number of words retained after processing his message, obtain content sender or the number of occurrences that his message contains. It can know whether it is accepted as relevant, whether it is rejected.

The trainee can, at any time, review all of his work in order to be able to correct his own errors in the future.

This Automated Learning Environment leads, at any time, to visualize, thanks to *Viewing* the different used available keywords, manually annotated contents, relevant contents, duplicate contents and replication contents.

It permits to work in Streaming or just by providing a message (especially for novices) in "Content Treatment".

In the case of Streaming, we have to specify, in *Listening / Monitoring*, whether it is from Twitter (Twitter is chosen optionally), from other Social Networks or from all the Web.

On the other hand, for the novices, one can enter the content, even update Keywords, in *Updating Files*. We can satisfy ourselves with an automatic (optional) or manual alert in *Alert*. With *File / New*, we start a new learning : just enter *Content File* and validate. We will still have this automated learning. We have *Data Mining*, *Neural Learning* and *Deep Learning*. *Data Mining* includes *Sequential Pattern Mining* , *Text Mining* and *Web Mining*, which are from the same family, according to

Bharanipriya et al.(2011) [162]. "Neural Learning" consists of "Relevant Information Retrieval" [14] and "Deep Learning" is still under development.

3.3.6 Discussion about the Automated Learning Environment

Interjections, blasphemies and emoticons are widely used by individuals to convey emotions such as anger, surprise, happiness, etc. To identify these three types of features, we used a combination of part-of-speech (POS) tags in the tagger, compiled lists of interjections and blasphemies on the Web and regular expression patterns for emoticons [14].

Messages from Algeria's target regions are checked for frequency bursts of disaster keywords and processed to identify evidence of a disaster. The advantage of our Automated Learning Platform is that it relies on evidence from social networks' first-hand reports. This multi-view template associates an input document vector with a multi-view space, based on neural network, whose contextual information is available.

Let $\hat{H}$ be the non-empty set ($\hat{H} \neq \emptyset$) of the terms that we have compiled (for example, terrorist act, explosion, earthquake, etc.) that make up the set of keywords or hashtags for natural and anthropogenic disasters.

Let $e_i \in C_n$, with $i \in [1, N]$, the incorporation of a content of the source message i relevant for, at least, a keyword or a hashtag $h_j \in \hat{H}$ with $j \in [1, M]$. C_n is the set of n contents.

We want to learn a generic space with the neural network

$$\hat{E} = \{e_k \text{ with } k \in [1, K]\} \quad (3.3)$$

which normalizes the differences :

$$\hat{E} = [E - \hat{R} - \hat{D}] \quad (3.4)$$

where : $\hat{R}$ is the set of retweets, $\hat{D}$ is the set of duplicate contents, and helps in better recovery of target content.

Thanks to the neural network, the transformation of e_i into e_k can be explained by :

$$e_i \rightarrow e_k = \{e_i, e_i \text{ is relevant for } h_j, e_i \notin \hat{R}, e_i \notin \hat{D}, h_j \in \hat{H}, j \in [1, M]\} \quad (3.5)$$

where $\hat{D}$ and $\hat{R}$ represent respectively the set of duplicate contents and re-tweets. The objective is then to maximize the size K of the set $\hat{E}$.

In order to show the effectiveness of our model, we examined a specific event - the

Boumerdes earthquake in May 21st, 2003 and the Algiers flood in November 10st, 2001 - and post-event messages on social networks.

Tweets were collected using the Twitter Search API and posts from other social networks were collected using the Radian6 tool. Both use the search keywords *Boumerdes* and *earthquake* for the Boumerdes earthquake and the *Algiers* and *flood* keywords for the Algiers floods.

Thanks to the neural network and after processing the content removing any inconsistencies such as punctuation, special characters, de-duplication and replication content [14], we have obtained figure 3.5 and figure 3.6, which present the number of relevant documents found for, respectively, each keyword or hashtag [14].

3.3.7 Performance Evaluation

Standard relevance queries and judgments, we identified a set of disaster-specific information needs, as proposed in [52]. It's a set of hashtags and keywords for Twitter and other social networks respectively. Data collection and filtering are at the heart of disaster management using social media. The algorithms used to warn and alert depend on the messages published on the social media and their quality. Thus, every effort must be made to maximize the number of contents relevant to the disaster and to remove non-informative messages as proposed in [132]. These contents are manually tagged (annotated) to remove those not related to the disaster (non-informative messages). Figures 7 and 8 present the number of relevant documents found for, respectively, each keyword or hashtag from the datasets.

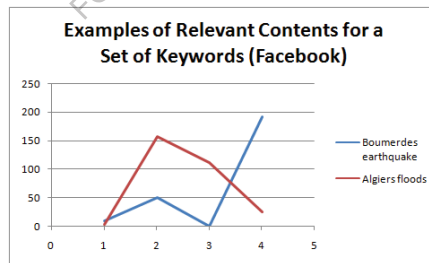


FIGURE 3.5 – Examples of Relevant Contents for a Set of Keywords

This content obtained, manually labeled (annotated) and cleaned from non-informative messages, will be used, in this space based neural network, to analyze live new messages through the learning allowed by this space based neural network, enriched by relevant messages and annotated manually.

The overall effectiveness of a large-scale disaster emergency response is a difficult

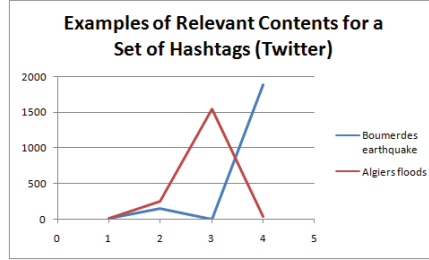


FIGURE 3.6 – Examples of Relevant Contents for a Set of Hashtags

entity to measure. Every disaster is different and every answer is different. In fact, similar events offer totally different results in their aftermath. For example, a category two hurricane can inflict much more damage than a category four, depending on the location of the impact. In addition, the response to an emergency rural and mountainous region of Kabylia, in Algeria, for example, will be totally different from the emergency response to a flood in capitals of major Algerian cities, as Algiers, Oran or Constantine. To effectively measure the overall response to these events, appropriate measures must be used and interpreted appropriately.

3.4 Conclusion

We have discussed in this paper everything related to an automated learning environment based on Deep Learning, namely Feedforward Neural Network. Concluding, we presented our Deep Learning-based automated learning environment, fundamentally personal, social, distributed, omnipresent, flexible, dynamic and suitable for disaster education. It is an extension of the real-time alert model used for the management of natural or anthropogenic disasters [14, 15]. This automated learning environment is based on a semi-supervised inductive technique of using abundant and unlabeled data, especially in times of crisis. This is just the beginning of this automated learning environment.

"Science is built up of facts, as a house is built of stones ; but an accumulation of facts is no more a science than a heap of stones is a house."

Henri Poincaré

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Chapitre 4

A Hybrid of Deep CNN-LSTM Model to improve Warning, Situational Awareness and Education in Managing Emergency : Case study of Covid-19

Summary

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4.1 Introduction

The volume of network and Internet traffic is expanding daily, with data being created at the zettabyte to petabyte scale at an exceptionally high rate. These can be characterized as big data, because they are large in volume, variety, velocity, and veracity.

Using social networking in crisis situations for sharing information in a timely manner [109] has become a standard practice in recent years. With the proliferation

of social media, an ongoing event [44] is being discussed on all these channels. There are generally qualitative differences [15, 16] in the information obtained from different sources. To get a complete view of the event, it is important to retrieve contents [16] from multiple sources. However, the challenge facing disaster managers is overwhelming when it comes to retrieving information shared on social media [74], with fine, excellent and net situational knowledge.

Several automated systems [15, 16, 98] have been designed to help disaster managers identify and filter useful information posted on social networking sites. Most of the work, has focused on using a few of social networks (if not the only social network, namely Twitter) as a source of information, only, on a few disaster management phases citeOlteanu2015,Albuquerque2015,Vulic2015, but few are, concurrently, dedicated to warning [15], crisis education [16] and situational awareness. The design of emergency management systems using different and diverse sources of information (all the Web), and dedicated primarily to warning, situational awareness and disaster education, is a challenge.

Deep learning models, recently applied to large-scale big data analysis, have shown remarkable performance in general [14, 15, 16]. This research work proposes a hybrid deep learning model to efficiently detect network intrusions based on a convolutional neural network (CNN) and a long short-term memory (LSTM) network. We use the deep CNN to extract meaningful features from big data and LSTM to retain long-term dependencies among extracted features to prevent overfitting on recurrent connections. The proposed hybrid method was compared with traditional approaches in terms of performance on a publicly available dataset, demonstrating its satisfactory performance.

This new approach allows integrating artificial intelligence technologies, deep learning and social media, in the Emergency Management Model [17, 93, 164]. It is based on an extension of the Recurrent Neural Network [89] of our previous approach, which is based on the Feedforward Neural Network of our previous approaches [16] used for disaster management and smart disaster education : this experience forms a background for this model. It combines representation training with alert, situational awareness and disaster education, while integrating encapsulations from multiple sources and retrieving information by combining multiple search results, providing some good ideas for that we have extended to improve Emergency Management.

In this research work [69], we try to identify relevant content dealing with impending catastrophic events. Once this information is retrieved (cleaned of non-informative information), it can be used to update disaster information (warning, situational awareness or disaster education) of disaster managers to make quick and effective decisions that could help people in need or to save lives. Thus, we provide, not only, a solution

to this challenge, but also, we achieve promising results.

Our study has four-fold main contributions.

1. We develop a Hybrid Deep Convolutional LSTM neural network-based model that uses content learning capabilities of multiple sources (all the Web) to automatically and efficiently capture real-time situation awareness reports distributed during large scale catastrophic events, using low-level content learning capabilities to automatically separate relevant content from non-informative information.
2. Using a dataset of keywords/hashtags related to various catastrophic events whether natural or anthropogenic, this model collects, according to their lexical similarity, relevant contents relating to various catastrophic events.
3. Keeping in mind the limitations of the previous work [14, 15, 16, 89, 97], we develop an event-independent model that can be used directly to filter content on multiple source at a time in future events. Experiments on multi-disaster-related contents flows with diverse characteristics show that our proposed model outperforms all the others.
4. Finally, we tested it immediately on the Global Corona Virus Pandemic, Called Covid-19, since January 2020, until nowadays. Then, we conclude and give some future works.

4.2 Proposed Emergency Management Model

Online messages contain important information [81] that can also be helpful in making quick decisions to help the affected community if they are dealt with quickly and effectively. Many types of processing techniques ranging from comparable document-aligned data [98], statistical analysis [44], natural language processing [165] to machine learning [15, 16, 109] to computational linguistics [45] have been developed for different purposes, without, fully exploiting this data, despite the existence of some resources, such as annotated data and standardized lexical resources.

Most event detection methods are based on keywords/hashtags used in tweets during catastrophic events to classify messages as real-time event reports, using a support vector machine (SVM).

Rogstadius et al., (2013) [72] were able to capture distributed situation awareness reports based on Twitter activity during natural disasters. Table 1 gives an overview of recent natural and anthropogenic disasters, and all their damage assessment.

Covid-19 is a pandemic of an evolving infectious disease. It first appears in Wuhan, China, in November 2019 and spreads then across the world. World Health Organization (WHO) asked for essential protective measures to prevent the saturation of

intensive care services and strengthen preventive hygiene. This global pandemic has prompted the cancellation of many sporting and cultural events around the world, the adoption of containment measures by several countries to postpone the creation of new centers of contagion, the closing of several countries' borders, and a stock market crash as a result of the uncertainty and concerns it has created for the global economy. It also has effects in terms of social and economic instability and is the pretext for the online dissemination of erroneous or conspiracy theory information. Luckily, with approximately 2 percent of the cases detected, the provisional death rate is lower than in previous coronaVirus epidemics. About roughly 110,270,288 cumulative cases were confirmed globally as of February 19, 2021, including 62,077,509 individuals healed and 2,439,834 dead. The contaminations number with the CoVid-19 corona Virus continues to increase to this day.

More than 4000 variants of SARS-CoV-2 have been identified around the world : a natural process as the virus acquires mutations over time to ensure its survival. British variant or B.1.1.7 (called VOC 202012/01 or B.1.1.7) : 64% more lethal.

First appearing in September 2020 in the UK, the UK variant was reported by UK authorities on December 14, 2020 and has since caused a sharp increase in cases on the island. It is not only more contagious but also 64% more deadly than the classic coronavirus.

South African variant (Called 501Y.V2) : 50% more transmissible. It is 1.5 times more contagious than SARS-COV-2, but not more lethal. Apparently, it seems to reduce the effectiveness of vaccines, in particular because of the E484K mutation. However, it tends to show that people infected with the South African variant of the new coronavirus have better immunity to other mutations in the virus.

Two Brazilian variants from the Amazon : the first variant, B1.1.248, was detected in early January in Japan in a family from Brazil and more specifically from the Amazon. This variant is more contagious. A second variant, called P.1, has appeared (one of the 18 variants of the coronavirus that circulated in the state of Amazonas). It is also said to be 1.1 to 1.8 times more likely to cause death.

TABLE 4.1 – Latest catastrophic events.

No	Catastrophic Event	Period	Damage
1	Volcanic eruption of Tambora	1815	92,000
2	China Floods	1931	200,000
3	Avalanche of Mount Huascarán	1970	75,000
4	Forest fire Haiti	Oct 2007	230,000 & 220,000
5	Tsunami in the Indian Ocean	2004	250,000
6	Haiti earthquake	2010	200,000

We suggest a new emergency management model focused on recurrent neural networks for warning, situational awareness, and education on social networks in this paper. It is based on an extension of Recurrent neural network [89] of our previous approach used to improve Fraud Detection and Time Series Forecasting. This last work is based on the Feedforward [15, 16] used in our previous works for disaster management and disaster education. This experience forms a background for this Emergency Management Model, based on RNN trained with LSTM used because of its ability to learn long term dependencies. This Emergency management model combines representation training with warning, situational awareness and disaster education, while integrating encapsulations from multiple sources and retrieving information by combining multiple search results (all the Web).

TABLE 4.2 – Overview of the most important floods and their damage (deaths) since 2001 in Algeria.

Situational Awareness	Identification Methods
Approaches	
[72]	CrisisTracker : Crowdsourced Twitter curation for disaster awareness
[74]	Summarization with social-temporal context
[75]	Building a Tweet Summarization Dataset Using a TREC Track
[76]	Semi-automated artificial intelligence-based classifier for Disaster Response
[79]	Multiscale analysis of Twitter activity before, during, and after Hurricane Sandy
Our New approach	Based on Hybrid of Deep CNN-LSTM

Whether it's Twitter, Facebook, Viber, Messenger, any forum or anything in the Web, these are platforms where people often express emotions. The data available on social networks differs in many ways from other Web sources (press articles, for example). These messages use less formal language, may contain words from more than one language, may have various grammatical and spelling errors, and are, for the most part, unstructured, fuzzy and short-lived [15, 16]. Their length and content vary considerably [88]. We detected emotions using features, such as : interjections, blasphemy, emoticons and the general feeling of the message, widely used by individuals to convey emotions such as danger, surprise, happiness, etc. To identify these features, we used a combination of par-of-speech (POS) tags, compiled lists of interjections and blasphemies on the Web and regular expression patterns for emoticons.

Our emergency management model based on a real-time recurrent neural network is best suited to situational awareness (see table 2). It uses multi-source content (from all the Web) (see table 3) learning capabilities to automatically and efficiently capture distributed real-time situational awareness reports during large-scale catastrophic events, using keywords/hashtags and tagged content. It collects the messages according to their lexical similarity, related to various catastrophic events, using disaster

TABLE 4.3 – Comparative table of all techniques and methods used in Models including our approach.

Ref	Identification Methods	Used OSN
[70]	Flood Disaster Game-based Learning	Twitter
[71]	Educational Purposes among the Faculty of Higher Education with Special Reference	Twitter
[74]	Summarization with social-temporal context	Twitter
[75]	Capitalizing on a TREC Track to Build a Tweet Summarization Dataset	Twitter
[76]	Semi-automated artificial intelligence-based classifier for Disaster Response	Twitter
[15]	Based on Artificial Neural Network (ANN)	Twitter/ Facebook
[16]	Based on FeedForward Neural Network (FFNN)	All the Web
[89]	Based on Recurrent Neural Network (RNN)	All the Web
Our New approach	Based on a Hybrid of Deep Convolutional LSTM (ConvLSTM)	All the Web

education (see table 4).

TABLE 4.4 – Comparative table of all techniques and methods used in Models with education including our approach.

Disaster Education Approach	Identification Methods
[70]	Flood Disaster Game-based Learning
[71]	Educational Purposes among the Faculty of Higher Education with Special Reference
Our Previous approach [16]	Based on FeedForward Neural Network (FFNN)
Our New approach	Based on Hybrid of Deep CNN-LSTM

Contents were collected from all online channels tracked automatically by the Online Listening Tool, namely Radian6 [88] from websites to all social media, such. Actually, many networking platforms allow access to their data via Application Programming Interface (API) [88]. Online listening tools provide the model, which reasonably represents the essentials, namely : harvesting contents, cleaning the data of non-informative information, enabling relevance thanks with the learning corpus obtained thanks to the tagged messages, and analyzing the results (verification and analysis of the results carried out in order to ensure adequacy to build disaster information such as warning, situational awareness and/or disaster education).

We present our new network model, ConvLSTM. The LSTM layer has been shown to be powerful in handling temporal correlation (too much redundancy for spatial data), its extension (of LSTM) has convolutional structures in both input-state and state-to-state transitions will solve this problem [166]. By stacking several ConvLSTM layers and forming a coding-prediction structure, we come to construct a

network model for these spatiotemporal sequence prediction problems.

The goal of crisis forecasting is to use the previously observed social networking sequence to forecast an event in a local region (for instance, Algiers, London, or Paris). From a machine learning perspective, this problem can be thought of as a spatiotemporal sequence prediction problem.

Suppose we observe a dynamic system represented by an $M \times N$ grid made up of M rows and N columns. Within each cell of the grid, there are P measures (word, bias) which vary over time. Thus, the observation at any time can be represented by a tensor X belongs to $R_{p \times M \times N}$, where R denotes the domain of observed traits. If we record the observations periodically, we will obtain a sequence of tensors

$$X_1, X_2, \dots, X_t.$$

The problem with spatiotemporal sequence prediction is to predict the most probable length- K sequence in the future given the previous J observations which include the current sequence :

$$\begin{aligned} \hat{Y}_{t+1}, \dots, \hat{Y}_{t+K} = \operatorname{argmax}_{\mathbf{x}_{t+1}, \dots, \mathbf{x}_{t+K}} \\ p(X_{t+1}, \dots, X_{t+K} | Y_{t-J+1}, Y_{t-J+2}, \dots, Y_t) \end{aligned} \quad (4.1)$$

For crisis forecasting, the observation at each time stamp is a 2D map. If we divide the map into non-tiled, non-overlapping patches and visualize the pixels inside a patch as its measurements, the problem is naturally a spatiotemporal sequence prediction problem. We note that our spatiotemporal sequence prediction problem is different from the one-step time series prediction problem because the prediction target of our problem is a sequence that contains both spatial and temporal structures.

The input to the network is a content e , as :

$$e = (w_1, w_2, \dots, w_t) \quad (4.2)$$

containing words $w_i \in \mathbf{W}$ each coming from a finite vocabulary V . $\mathbf{V}$ is the set of contents issued from the social media.

How to determine the coefficients $w_0, w_1, \dots, w_n$? To the Error feature : if $y = 1$, we want $p(x)$ to be as big as possible. We define the error in this case by :

$$-\ln(p(x)) \quad (4.3)$$

Symmetrically speaking, we want $p(x)$ as small as possible if $y = 0$. The error is

then :

$$-\ln(1 - p(x)) \quad (4.4)$$

So, the general formula is :

$$error = -y * \ln(p(x)) - (1 - y) * \ln(1 - p(x)) \quad (4.5)$$

Once an error function has been defined, the problem of learning becomes an optimization problem : find the coefficient vector w^* which minimizes the error. In the case of logistic regression, this vector is unique because the error function is convex.

Once the optimum w^* coefficient vector is determined, a program (classifier) is available to classify a new individual. To estimate the error probability of the classifier, it is necessary to have an independent test set.

CNNs are regularized variants of multilayer perceptrons, usually meaning totally linked networks, where each neuron in one layer is linked to the next layer [17, 18, 47, 93, 142, 164].

The *fully-connectedness* of these networks makes them susceptible to overfitting information, where traditional methods of regularization, include adding to the loss function, some form of magnitude measurement of weights.

$$\forall n \in [1, 2, n_C^{[l]} \\ conv(a^{[l-1]}, K^{(n)})_{x,y} = \quad (4.6)$$

$$\psi^{[l]}(\sum_{i=1}^{n_H^{[l-1]}} \sum_{j=1}^{n_W^{[l-1]}} \sum_{k=1}^{n_C^{[l-1]}} K_{i,j,k}^{(n)} * a_{x+i-1,y+j-1,k}^{[l-1]} * b_n^{[l]}) \quad (4.7)$$

$$dim(conv(a^{[l-1]}, K^{(n)})) = (n_H^{[l]}, n_W^{[l]})$$

Although CNNs take a different approach to regularization. Convolutionary networks, inspired by biological processes, where the pattern of communication between neurons follows the organization of the visual cortex of the animal : individual cortical neurons respond to stimuli only in a small area of the visual field known as the receptive field (see Figure 1). CNNs use very little pre-processing : they learn the filters that were hand-engineered in conventional algorithms [47].

Artificial intelligence (AI) is a combination of reinforcement learning (RL) and deep learning (DL) [104], represented mathematically, as :

$$AI = RL + DL \quad (4.8)$$

where : AI represents Artificial Intelligence, RL represents Reinforcement Learning, and DL represents Deep Learning.

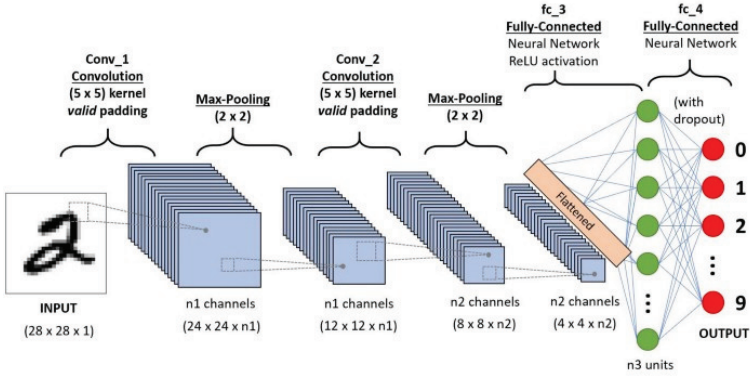


FIGURE 4.1 – The structure of Convolutional NN.

A feedforward neural network (FFNN) is an automated learning classification algorithm which is made up of organized in layers, similarly to human neuron processing units. Each unit in a layer (known as node) relates to all other units in these layers. These layers connections can have a various weight measuring the potential amount of the network knowledge. In the network, information processing requires data entry from the input units and flows through the network, flowing from one layer to the other before the output units. If the neural network operates normally (as a classifier), then there will be no feedback [104] between layers. Logically, FFNN can handle tasks based on first come first serve input bases. As for the feed-backward NN (FBNN), it can use internal state memory to process sequence of data inputs, such as Recurrent Neural Network (RNN).

As a class of feedforward neural networks, recurrent neural networks (RNNs) are augmented by the inclusion of recurrent edges connecting adjacent time steps. Figure 1 shows the well-known Elman, recurrent neural network [138], according to Jordan's original idea [167].

The formula for the current state is

$$h_t = f(h_{t-1}, x_t) \quad (4.9)$$

Applying activation function $\tanh$:

$$h_t = \tanh(\sigma_{hh}.h_{t-1} + \sigma_{xh}.x_t) \quad (4.10)$$

σ is weight, h is the single hidden vector, σ_{hh} is the previous hidden state weight,

TABLE 4.5 – Overview of the coronavirus pandemic (Covid'19) and their damage (affected, dead and healed) since May 31st, 2020, in Algeria.

Period	Affected	Death	Healed
May 31 st , 2020	9,394	653	
June 06 th , 2020	10,05	698	
July 21 st , 2020	24,278	1,1	
August 24 th , 2020	41,858	1,446	
October 1 st , 2020	52,658	1,783	36,958
November 10 th , 2020	63,446	2,077	42,626
November 12 th , 2020	64,257	2,093	42,980
March 16 th , 2021	115,410	3,040	79,994

σ^{xh} is the current input state weight, $\tanh$ is the funtion of Activation, that introduces a Non-linearity squashing the activations to the range $[-1,1]$. Output :

$$y_t = \sigma_{hy} \cdot h_t \quad (4.11)$$

y_t is the output state. σ_{hy} is the weight at the output state.

According to Figure 1, we can use two equations to describe this type of LSTM, all calculations necessary for computation at each time step on the forward pass is :

$$h_t = \alpha(\sigma_{hx} \cdot x_t + \sigma_{hh} \cdot h_{t-1} + b_h) y_t = \beta(\sigma_{yh} \cdot h_t + b_y) \quad (4.12)$$

where the σ terms denote weight matrices (e.g. σ_{hx} is the matrix of weights between the input and hidden layers).

The b terms denote bias vectors (e.g. b_h is hidden bias vector) which allow each node to learn an offset.

α denotes the hidden layer function.

In general, α is usually an element-wise application of a sigmoid function and β is the output layer function.

LSTM referred to a standard kind of neural network which extended over time, with edges, that feed into the next time step, rather than feeding into the next layer. It is constructed to sequences recognition, for instance, a text or a speech signal. It has cycles indicating short-memory presence in the net. LSTM is like a hierarchical network where the input need processing hierarchically in the form of a tree because there is no time to input sequence.

It has achieved remarkable success in sequential learning problems.

Although RNN is not deep in space, it is inherently deep in time since each hidden state is function of all previous states [138].

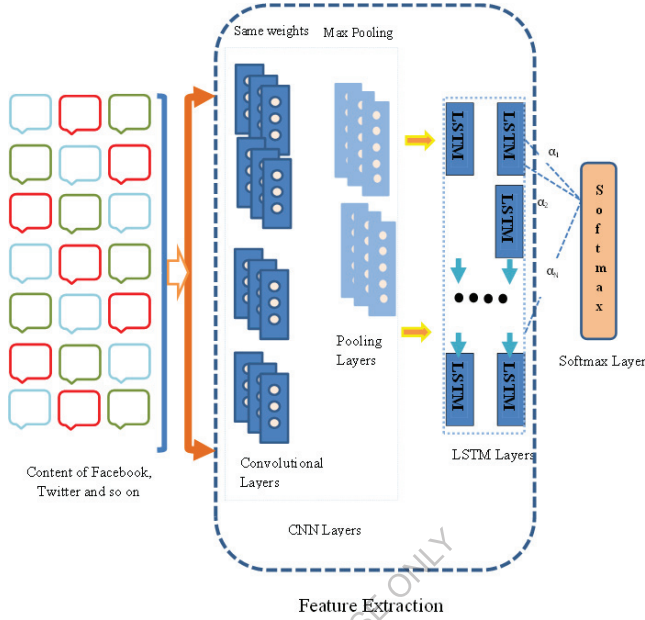


FIGURE 4.3 – Overview of the A Hybrid CNN-LSTM cell Architectures

2. Forget gate - using sigmoid, find information to delete from the block. It analyses the previous state h_{t-1} and material input x_t for each number in cell state c_{t-1} , and selects the number from 0 (omitting it) and 1 (keeping it)..

$$f_t = \sigma(\omega_{fx} * x_t + \omega_{fh} * h_{t-1} + b_f) \quad (4.15)$$

3. Output gate - To select the output, the input and block memory are used. The Sigmoid function selects values to pass 0,1 and the Tanh function gives weight to the values transferred, evaluating their degree of significance varying from -1 to 1 and multiplied by the Sigmoid output.

$$j_t = \sigma(\omega_{jx} * x_t + \omega_{jh} * h_{t-1} + b_j) \quad (4.16)$$

$$h_t = j_t \odot \tanh(c_t) \quad (4.17)$$

where

$\odot$ denotes element-wise multiplication. ω is the logistic sigmoid function. i, f

and j are respectively the input gate, forget gate and output gate. c is the cell activation vectors, all of which are in the same size as the hidden vector h in level k .

The entire RNN LSTM modeling procedure has been studied with the aim of introducing systematic methods leading to always efficient models, namely the collection of learning, preprocessing and post-processing of data, various types of initializing weights, algorithms for learning, activation and error functions. Although all of these factors affect its performance, increased attention has been focused on finding the best architecture.

4.2.1 Foundation of neural learning

We have a recurrent neural network with hidden layers taking as input contents as : $e = (w_1, ..., w_i, ..., w_n)$ contains words each coming from a finite vocabulary.

Let :

$$e_i \in C_n, \text{ with } i \in [1, N] \text{ and } e_i = (w_{i1}, w_{i2}, ..., w_{in}) \quad (4.18)$$

containing words each coming from each a finite vocabulary V , the incorporation of a content of the source message i relevant for, at least, a keyword or a hashtag :

$$H_j \in \mathbf{H} \text{ with } j \in [1, M] \quad (4.19)$$

We want the learning of a generic space with the neural network, as :

$$\mathbf{E} = \{e_k; \quad k \in [1, K]\} \quad (4.20)$$

which normalizes the differences :

$$E = [\mathbf{E} - \mathbf{RDF}] \quad (4.21)$$

where :

$$\mathbf{RDF} = [\mathbf{R} + \mathbf{D} + \mathbf{F}] \quad (4.22)$$

Thanks to the Hybrid of Deep Convolutional neural network-LSTM, the transformation of e_i into e_k can be explained by :

$$e_i \rightarrow e_k = \{e_i; \text{ such as } e_i \text{ is relevant for } H_j \text{ and } w_i;\} \quad (4.23)$$

$$H_j \in \mathbf{H} \text{ with } j \in [1, M]$$

where :

$$w_l \in \mathbf{W}; \text{with } l \in [1, L]; e_i \in [\mathbf{R} + \mathbf{D} + \mathbf{F}]; \quad (4.24)$$

$$\text{with } k \in [1, K]; i \in [1, N]$$

$\mathbf{D}$, $\mathbf{R}$ and $\mathbf{F}$ represent respectively the set of duplicate retweets, duplicate contents and false alerts.

$\mathbf{W}$ represents the set of words of a language that L is its dimension.

The objective is then to maximize the size K of the set E .

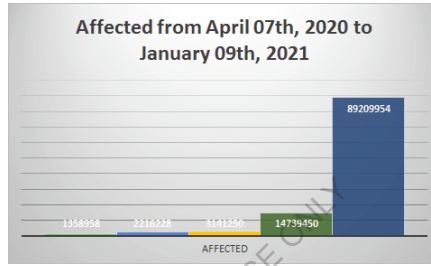


FIGURE 4.4 – Global assessment of the CoronaVirus Pandemic.

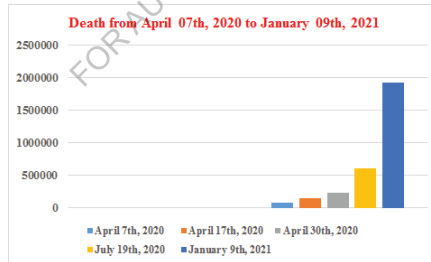


FIGURE 4.5 – Global assessment of the CoronaVirus Pandemic.

Social media data help respond to disasters [79]. During crisis events, citizens easily turn to social networks to confide in, quickly disseminate information and learn useful insights. Social media improves people's knowledge of the situation, facilitates the dissemination of information (especially in emergencies), enables to learn useful insights, early warning systems and helps coordinate relief efforts. In addition, the spatio-temporal dissemination of messages relating to crises facilitates real-time monitoring and evaluation of this disaster, before, during and after events [79].

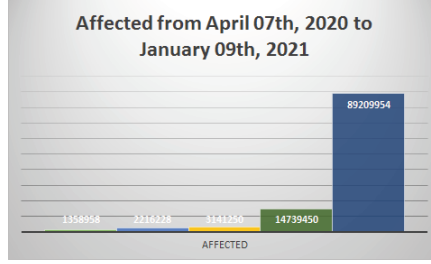


FIGURE 4.6 – Global assessment of the CoronaVirus Pandemic.

Most disaster publications assume that the media are the most important mitigation tool for managers because their content creates awareness of disasters and risks. Victims, volunteers and relief organizations are increasingly using social media to report and take action on high profile events [72]. Researchers show a correlation between per capita social media activity and disaster damage, making it easy to quickly assess [79].

4.2.2 Warning and Alert

The training data is created from the Coronavirus pandemic, Covid-19. This information, easily obtained using the neural network, is manually annotated by volunteers.

4.2.3 Situational Awareness

In crisis situations, the essential decision making needed depends heavily on the availability, quality, and timeliness of relevant information available to decision-makers. Our approach in designing situational awareness systems is to design warning model that consider situations and events as fundamental entities. An important aspect of emergency situation awareness using social media consists to detect and characterize the emergency-related event while it is not still known (we will be the first to know). Thus, we will be better equipped to take all the precautions and the luck on our side. In the evolving pandemic, knowing and especially applying, first of all, wearing the bib and making physical/social distance, will play a significant role in saving lives. This will serve as first operations, among others, to apply. Physical/social distancing is significantly influenced by situational awareness and disaster education [170]. Thus, it can be inferred that increasing situational awareness, in times of public health crisis, using disaster education, can significantly increase the protective health behavior adoption and contain the infectious diseases spread. (See Figures 5, 6 and 7).

4.2.4 Disaster Education

This Emergency Management Model is designed to support an introductory traineeship in emergency management for citizens, trainees and future disaster managers. Thus, the trainee can use this tool in three modes [16]. Novice Mode permits him to use a complete set of automated design and learning tools, such as observing of various programs at work, experimenting them and gradually learning from his experience, observations and mistakes. Beginners Mode permits him, at any point, to ask this tool to generate (move on) the next step. This tool analyzes knowledge and provides both the optimal stage and a list of all relevant operations. If he is not satisfied with the optimal operation suggested by the system, he can choose any appropriate operation using adaptive hierarchical menus. In the Online manual rehearsal Mode, at any time during the work, the trainee has a menu to access all previous courses : presentation of any previously learned concept, demonstration of all the examples learned and analysis of any problem explained or resolved. This mode provides access to the material learned from the course as a reference, thereby supporting example-based online help.

Educational messages [170] play a role in improving situational awareness in times of public health emergencies.

The Covid-19 pandemic is an emerging infectious disease [171] caused by the coronavirus SARS-CoV-2.

Disaster education for the Covid-19 pandemic involves consists of advising to always strengthen preventive hygiene, namely elimination of physical contact, kisses and handshakes, coughing and/or sneezing into the crook of the elbow, using disposable tissues, taking physical/social distancing, wearing a bib, end of gatherings and major events as well as unnecessary trips, promotion of hand washing and avoiding any social or cultural regrouping. But above all, this disaster education consists of constantly rehashing this advice on all information channels, websites and all social and networking media to have as much situational awareness as possible.

By adapting educational programs designed by many academic institutions, hospitals, professional organizations, governments, and non-governmental organizations, this Emergency Model is also designed to support an introductory traineeship to prepare the health system and health personnel to meet the health needs of populations affected. It consists of Standardizing good practices by developing the *core competencies* of essential knowledge and skills for disaster health workers.

Hundreds of competencies for disaster healthcare personnel have been developed and endorsed by governmental and professional organizations and societies [172], namely :

- competencies tailored to a given position or function during a disaster ;
- unique competencies focused on skill level rather than role or function ;

- competencies based on specific roles as well as proficiency levels;
- competencies based on specific roles as well as proficiency levels;
- graded emergency nursing skills according to the stages of the disaster management process;
- specified competencies as *core* competencies for various target groups, and
- *cross-cutting* competencies applicable to all healthcare workers.

Unfortunately, imprecise and inconsistent terminology and structure are evident throughout the reviewed competency sets. It is necessary of Universal acceptance and application of these competencies. The aim of this approach is developing a framework and standardized terminology for the articulation of competency sets for disaster health professionals accepted and adapted universally.

4.2.5 Performance Evaluation

Standard relevance queries and judgments. The objective is then to maximize the size K of the set E_K . The training data is created from the Coronavirus pandemic, Covid-19. This information, easily obtained using the neural network, is manually annotated by volunteers. In order to show the effectiveness of our model, we examined specific events of Covid-19 and post-event messages on several social media : Twitter, Facebook, LinkedIn, Instagram, Google+, Youtube, Messenger, Viber and so on.

All experiments reported here were performed on this set of data. With our new proposed model and after cleaning and de-duplication, we have the following data set summarized in table 4.6 and figure 4.2.5.

TABLE 4.6 – Examples of Relevant Content of Global Corona Virus Pandemic (Covid-19) for a Set of Hashtags and Keywords for all social networks.

Model	Global Corona Virus Pandemic (Covid-19)
[15] (Neural Network)	15,619
[16] (FeedForward NN)	16,942
[89] (Recurrent NN)	17,846
Our New Approach (ConvLSTM)	19,925

We identified a set of disaster-specific information needs. It's a set of keywords. Data collection and filtering are at the heart of disaster management using social media. Every effort should be made to increase the number of relevant contents while removing non-informative data (messages manually annotated to cleaning them).

To refine the results, generally, different topics behave better with different functions and keywords/hashtags. The goal was to identify all relevant content relative to the predefined keywords. The overall effectiveness of a large-scale disaster emergency

response is a difficult entity to measure. Every disaster is different and every answer is different.

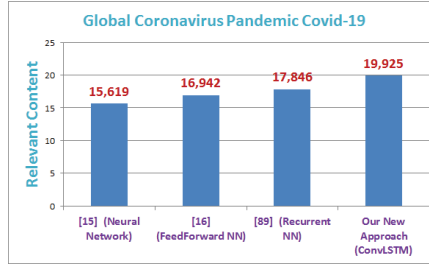


FIGURE 4.7 – Examples of Relevant Content of Global Corona Virus Pandemic (Covid-19) for a Set of Hashtags and Keywords for all social networks.

In fact, similar events offer totally different results in their aftermath. For instance, a category four flood in Algiers, Algeria's capital, will cause much more damage than a category two flood in a smaller town. In addition, the response to an emergency in rural region is totally different from the emergency response to Bouira earthquake, in a regional city, according to Brown et al. (2005) [173]. Appropriate indicators must be used and interpreted correctly to accurately assess the overall response to these incidents. Although there are many theories, it is very difficult to evaluate the effectiveness of a method because of the inherent irregularity of disasters. Two disasters are not alike and it is therefore difficult to determine if one method is better than another. In a real disaster, the answer does not have to be a test of new ideas with inherent confusion, but rather a well-tested and realized answer [173]. Using social networks' disaster datasets leads to vague words, contradictory and incomplete details, and general uncertainty. When modeling and simulation are necessary for a cheap and time-efficient system observation method, using disaster data is quick and difficult at best and also for testing several inputs and different results. One of the best reasons to implement a simulation for an observer is drawing the benefits of simulation in terms of cost and time [173]. Otherwise, the pilot alert system presented here consists of validating the research bases carried out as part of this work.

4.3 Conclusion

This work is based on extracting, in real time, information on catastrophic events from multiple sources (*on the Web* and to immediately alert disaster managers so that they can make quick and effective decisions that could, perhaps, save lives. Indeed, we

propose a new ad hoc real-time alert model for the management of disaster, whether natural or anthropogenic, based on a new multi-view recovery model from multiple sources. Such an approach is really useful for local disaster monitoring, but also for helping to make appropriate decisions.

1. We have developed a neuron-based model that uses low-level content learning capabilities to automatically separate relevant information from redundant (*eg. retweets for Twitter*) or abusive information (*at the level of the community, for example*).
2. This model, based on a Hybrid Deep Convolutional LSTM, uses the learning capabilities of multiple source content (*all the Web*) to automatically retrieve relevant information using a set of keywords and hashtags respectively, related to various catastrophic events, whether natural or anthropogenic.
3. Keeping in mind the limitations of the previous work [14, 15, 16, 89], we have developed an event-independent alert model that can be used directly to filter content from multiple sources at future events. Experiments on multi-disaster-related content flows with diverse characteristics show that our proposed model outperforms views in previous approaches. While our approach filters content (*Twitter tweets and Facebook posts at once*).
4. Once we developed this real-time alert model, we immediately tested it on the actually Corona virus Pandemic (Covid-19).
5. We have also proposed a warning network propagation scheme that will enable disaster managers to make effective decisions quickly and effectively to save lives.

"A scientist can be productive in various ways. One is having the ability to plan and carry out experiments, but the other is having the ability to formulate new ideas, which can be about what experiments can be carried out by making (the) proper calculations. Individual scientists who are successful in their work are successful for different reasons."

Linus Pauling

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General Conclusion & Perspectives

This chapter closes this thesis report which deals with the consideration of real-time disaster management using social networks and in particular machine learning to extract new relevant messages that arrive in social networks. Throughout this report, we have presented disasters, disaster management, the different models of disaster management. We also presented social networks and monitoring tools for its content, which is continually updated by people. We have presented all classification techniques, from SVM to machine learning. Then we improved these classic feasibility conditions to extract relevant content, using artificial intelligence techniques and automated learning.

Summarizing Chapters

The state of the art of retrieving relevant contents of disaster from social media consists of the overview of disaster and disaster management, social networking and models of retrieving information from social networks : it is presented in the Chapter 1. First of all, this chapter recalls the modelisation of disaster, after presenting all its definitions. This mathematical modelisation of disaster and also disaster management models allow the researchers to have an idea of different aspects and concepts necessary to save lives.

In Chapter 2, we present our basic implementation of the first alert model based on automated neural network for extracting relevant content from Twitter and Facebook using keywords and hashtags of disaster. This Chapter allows us to begin designing and implementing the heart of the automated learning environment.

The chapter 3 represents the first enhancement of the automated learning environment : it is based on Deep Learning, namely Feedforward Neural Network, extract, in real time, information on catastrophic events from multiple sources (all the Web) and to immediately alert disaster managers so that they can make quick and efficient decisions that could save lives. Indeed, we propose a new ad hoc real-time alert model for the management of disaster, whether natural or anthropogenic, based on a new multi-view recovery model from multiple sources.

Chapter 4 presents an other improvement of our real-time automated learning

environment analysis that is based on a Hybrid of Deep Convolutional LSTM neural network.

Results obtained

To evaluate our proposed model for extracting relevant information through semantically-based learning, we need an annotation for a set of contents. We randomly sampled 1366 messages (*with 547 Facebook content and 819 tweets*) after deleting the duplicates. These messages were observed independently by ourselves [14, 15]. The goal was to identify all relevant content relative to the predefined keywords. This content manually labeled (*annotated*) and cleaned from non-informative messages, is used, in this neural network based space, to analyze live new messages through the learning allowed by this neural network based space, enriched by relevant messages annotated manually. As we want our alert model to be event - independent, it must be able to be used directly on content posted at later events. We therefore adopt the approach of using a set of features for the content extraction task. And since we have a large number of data sets (*14 sets of data*), a few are dedicated to training and others to check the performance of our proposed model.

In order to show the effectiveness of our model, we examined two specific events, namely the Boumerdes earthquake in May 21st, 2003 and the Algiers flood in November 10st, 2001 - and post-event messages on social networks. Tweets were collected using the Twitter search API and posts from other social networks were collected using the Radian6 tool. Both use the search keywords *Boumerdes* and *earthquake* for the Boumerdes earthquake and the keywords *Algiers* and *flood* for the Algiers floods. Thanks to the Feedforward neural network [16] and after processing the content by removing inconsistencies such as punctuation, special characters, deduplication and replication content [14, 15], we got results confirming the interest to improve this machine learning platform by moving from a simple Automated Neural Network (ANN) to the Feedforward Neural Network (FFNN) in Deep Learning. This encourages us to move forward in the task of improving it.

In Chapter 4, this Emergency Management Model [69] is designed to support an introductory traineeship in emergency management for citizens, trainees and future disaster managers. Thus, the trainee can use this tool in three modes [16] : Novice, Beginners, and Online manual rehearsal Mode.

Educational messages play a role in improving situational awareness in times of public health emergencies.

The Covid-19 pandemic is an emerging infectious disease caused by the coronavirus SARS-CoV-2. Disaster education for the Covid-19 pandemic involves consists

of advising to always strengthen preventive hygiene, namely elimination of physical contact, kisses and handshakes, coughing and/or sneezing into the crook of the elbow, using disposable tissues, taking physical/social distancing, wearing a bib, end of gatherings and major events as well as unnecessary trips, promotion of hand washing and avoiding any social or cultural regrouping. But above all, this disaster education consists of constantly rehashing this advice on all information channels, websites and all social media to have as much situational awareness as possible.

This Emergency Management Model [69] is designed to support an introductory traineeship in emergency management for citizens, trainees and future disaster managers. Educational messages play a role in improving situational awareness in times of public health emergencies. Training data is created from the Coronavirus pandemic, Covid-19. This information, easily obtained using the neural network, is annotated manually by volunteers. In order to show the effectiveness of our model, we looked at specific Covid-19 events and post-event posts on several social media : Twitter, Facebook, LinkedIn, Instagram, Google+, Youtube, Messenger, Viber, etc.

All of the experiments reported here were performed on this dataset. With our new model [69] and after cleaning and deduplication, we obtained satisfactory results, above all, after having compared them to the previous models, namely the model based on ANN [14, 15] we had 15,619, the model based on Feedforward neural network [16] we have 16,942 and finally the RNN model [89] we have 17,846.

Our work [69] consists of extracting relevant content not only from Twitter and Facebook, but also from all social media, including, Viber, Instagram, and so on. Thus, our challenge consists of, effectively, filling this lack of information. The present work is relevant for the following reasons. The model used is based on a Convolutional LSTM (ConvLSTM). It recovers in streaming (in real-time, from the social media, contents. Then, it cleans up them with eliminating, before all, non informatifs contents (duplicated and replication contents).

This model integrates encapsulations from multiple sources, and thus retrieves accurate information by combining multiple search results from multiple sources' all the Web). The annotation of the contents obtained by the initial learning is done by volunteers. This Emergency Management Model is also used for smart disaster education by dint of handling it, we discover basic elements of triggering disasters. As it is manipulated, one becomes familiar with certain basic concepts of all disasters. One of the greatest advantages of this model is, above all, used for all types of disasters, whether natural or anthropogenic.

Perspectives

As for future works, we believe that our study has explored many potential gaps in disaster management for the future, the main flaw of which is the neglect of the P2P concept. For pure improvements, we can start with :

1. An improvement can come from the P2P concepts which have to be integrated into Geo-collaborative applications since today's applications, based on the client / server paradigm (C / S), suffer from major drawbacks such as the need for direct, highly available and reliable communication channels. Indeed, during disasters, naturally representing highly dynamic mobile environments with frequent changes in topology (*network*) and potential disconnections. Due to the large autonomy of the peers involved leading to self-regulation behavior, such networks are very well prepared to address frequent changes in topology [60] such as information and positions. Peers can not only communicate with a server but also exchange information between them. In addition, the scalability feature is related to this by providing support for dynamically joining and leaving nodes and spontaneous networking. Thus, the benefits of applying the P2P paradigm [59] have to be applied to the Geo-Collaboration for Disaster Management as an alternative to the most common Client / Server approaches of GIS and propose to interconnect mobile operators through P2P networks. This P2P network does not require an Internet connection or a telecom service provider network to communicate with others. In addition, the Peer-to-Peer paradigm provides a unique opportunity for service offerings by individual users. Services based on local user proximity can benefit from Peer-to-Peer provisioning without infrastructure support.
2. This work can also be improved by generalizing the application of Peer-to-Peer to mobile environments which offers a number of opportunities and challenges since, originally, Peer-to-Peer was not designed for mobile environments. In addition to well-known file-sharing applications based on Peer-to-Peer networks, new wireless applications are also possible in mobile networks, especially for multi-hop links as in ad-hoc mobile networks.

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List of Publications

- International journal with reading committee :

Bouzidi, Z., Amad, M. and Boudries, A., (2019), Intelligent and Real-time Alert Model for Disaster Management based on Information retrieval from Multiple Sources, International Journal of Advanced Media and Communication, Vol. 7, No. 4, pp. 309-330, DOI : 10.1504/IJAMC.2019.111193

Bouzidi, Z., Boudries, A. and Amad, Mourad, (2019), *Deep LSTM-based Automated Learning Environment using Smart Data to Improve Awareness and Education in Time Series Forecasting*, MC Medical Sciences journal, Vol 1, Issue 5, Dec 2021, <https://doi.org/10.55162/MCMS.2021.01.034>

Bouzidi Z., Amad M. & Boudries A., (2022), *Enhancing Warning, Situational Awareness, Assessment and Education in Managing Emergency : Case study of Covid-19*, Springer Nature Computer Science, 2022, vol 1, No 454, doi :10.1007/s42979-022-01351-2

- International conferences with reading committee :

Bouzidi Z., Boudries A. and Amad M., (2018), A New Efficient Alert Model for Disaster Management, Proceedings of Conference AIAP'2018 : Artificial Intelligence and Its Applications, El-Oued, Algeria

Bouzidi, Z., Boudries, A. & Amad, M. (2020). Towards a Smart Interface-based Automated Learning Environment Through Social Media for Disaster Management and Smart Disaster Education. Advances in Intelligent Systems and Computing. SAI 2020. Vol 1228. Springer, Cham, 443-468. https://doi.org/10.1007/978-3-030-52249-0_31

Bouzidi, Z., Amad, M. and Boudries, A. (2020b), Deep Learning-based Automated

Learning Environment using Smart Data to improve Corporate Marketing, Business Strategies, Fraud Detection in Financial Services and Financial Time Series Forecasting. In International Conference on 'Managing Business through Web Analytics - (ICMBWA2020)', Khemis Miliana University, Algeria

Bouzidi, Z., Boudries, A. and Amad, M., (2021), Deep Learning and Social Media for Managing Disaster : Survey, Intelligent Systems Conference 2-3 September, 2021, (IntelliSys 2021), Amsterdam, The Netherlands, Accepted

Bouzidi, Z., Boudries, A. and Amad, Mourad, (2021b), LSTM-based automated learning with smart data to improve marketing fraud detection and financial forecasting, 5th International Scientific Conference on Economics and Management, (EMAN 2021), Serbia, Accepted

Bouzidi, Z., Boudries, A. and Amad, Mourad, (2021c), Enhancing Crisis Management because of Deep Learning, Big Data and Parallel Computing Environment : Survey, ID : 443, Proceedings of the 3rd International Conference on Electrical, Communication and Computer Engineering (ICECCE), 12-13 June 2021, Kuala Lumpur, Malaysia

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Abstract

Nowadays, technology is constantly evolving and more and more quickly. An event is discussed in Social networking. It must ensure, before deployment, that a first-hand message will always arrive to disaster managers, to take quick and efficient decisions that will save lives, and this notwithstanding its fake news. The real-time retrieving of relevant content from social media in crisis event is very important. This relentless development responds to the demand of the general public. Obviously, the integration of new functions, always more sophisticated, in our daily tools, makes their development more and more complex, in particular at the real time level.

Keywords : Disaster, Neural learning, Relevant content, Social media.

Résumé

De nos jours, la technologie évolue constamment et de plus en plus rapidement. Un événement est discuté dans les réseaux sociaux. Il doit s'assurer, avant le déploiement, qu'un message de première main arrivera toujours aux gestionnaires de catastrophes, pour prendre des décisions rapides et efficaces qui sauveront des vies, et ce notwithstanding son fausses nouvelles. La récupération en temps réel du contenu pertinent des médias sociaux en cas de crise est très importante. Ce développement incessant répond à la demande du grand public. Evidemment, l'intégration de nouvelles fonctions, toujours plus sophistiquées, dans nos outils quotidiens, rend leur développement de plus en plus complexe, notamment au niveau temps réel.

Mots-Clés : Catastrophe, Apprentissage neuronal, Contenu pertinent, Média sociaux.

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